

Chemistry

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1. What is a basic oxide (1138)

- A) MgO
- B) MgF₂
- C) CO₂
- D) CO
- E) HNO₃

2. Ice melting is a reaction; (686)

- A) Isoenthalpic
- B) Exothermic
- C) None of the above
- D) IsoEntropic

3. Given the system at equilibrium at temperature T (exothermic reaction $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ if the T decreases, it is observed that: (1567)

- A) Increases the speed of reaction
- B) Decreases the activation energy
- C) The equilibrium shifts to the right
- D) Increases activation energy
- E) The equilibrium shifts to the left

4. Which of the following triads of elements consists of a metal followed by a transition metal and by a non-metal? (340)

- A) P, Fe, O
- B) Na, Ba, P
- C) Li, Cu, Zn
- D) Ti, Fe, S
- E) Be, V, S

5. Which statement regarding a Daniel cell with frit is correct? (2098)

- A) All the others are wrong
- B) Cu⁺ ions cross the porous septum
- C) The mass of the electrode at the anode decreases during the discharge
- D) There are sulfide ions in solution
- E) Zinc ions pass through the porous septum

6. Which are the names of the following compounds? H_2SO_4 , Na_2SO_3 (1349)

- A) Sulfuric acid/sodium sulphate
- B) Sulfuric acid/sodium sulphite
- C) Sulphurous acid/sodium sulphate
- D) Sulphurous acid/sodium sulphite
- E) None of the above

7. Which of these is the correct electron configuration of titanium? (1166)

- A) $[\text{Ar}] 3d^2 4s^2$
- B) $[\text{Ar}] 4d^2 4s^2$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^2$

8. Which of these elements has the greatest electron affinity? (1354)

- A) H
- B) He
- C) F
- D) Approx
- E) C

9. Which of these elements has the lowest first ionization energy? (101)

- A) Hydrogen
- B) Fluorine
- C) Helium
- D) Magnesium
- E) Sodium

10. What is the freezing point of a weak electrolyte? (1173)

- A) 273 K
- B) Above 0°C
- C) Between 273K and 373K
- D) Less than 273K
- E) None of the above

11. What do benzene and ethene have in common? (1712)

- A) sp^2 hybridization
- B) resonance
- C) they oxidize easily

12. compared to a well-defined salt. a base is an entity that: (413)

- A) accepts a proton
- B) donates a proton
- C) no right answer
- D) accepts an electron
- E) donate an electron

13. Indicate the valence of nitrogen in each of the following compounds: NO₂, Pb(NO₂)₂, N₂O₅ (1193)

- A) 4, 3, 3
- B) 4, 5, 3
- C) 4, 3, 5
- D) 2, 3, 5
- E) 2, 5, 5

14. the ratio of the energy of a photon to the frequency of radiation is equal: (233)

- A) to the wavelength
- B) to Planck's constant
- C) a pigreco
- D) at the speed of light
- E) at the temperature

15. minimum number of carbon atoms to have a structural isomer? (438)

- A) 1
- B) 2
- C) 3
- D) 4

16. Which of the following compounds contains a hydroxyl group (-OH) (85)

- A) Alcohols
- B) ethers
- C) esters
- D) cycloalkanes
- E) alkenes

- 17.** An exothermic reaction with 5 moles of gas in the reactants and 4 moles in the products, what change induces an increase in the concentration of products? (1250)
- A) Pressure increase
 - B) Volume increase
 - C) Increase T
 - D) Addition of reagents
 - E) Adding products
- 18.** A reaction is endothermic if (13)
- A) $\Delta H > 0$
 - B) $\Delta H < 0$
 - C) $\Delta E < 0$
 - D) $\Delta S > 0$
 - E) $\Delta G = 0$
- 19.** Which of these statements about methanol is correct? (240)
- A) Its molecular mass is 32
 - B) Contains a carboxyl group
 - C) It is an ester
- 20.** What determines the spontaneity of a reaction (2461)
- A) temperature
 - B) ΔG
 - C) ΔS
- 21.** Heisenberg's Uncertainty Principle (661)
- A) It is not possible to know precisely the position of an electron
 - B) It is not possible to know precisely the position and the momentum of at the same time an electron
 - C) It is not possible to know precisely the speed of an electron
- 22.** How many molecules are there in 0.5 moles of CO_2 (255)
- A) $0.5 \cdot N_A$
 - B) $3 \cdot N_A$
 - C) 1
 - D) $1.5 \cdot N_A$

23. Which of the following pairs of elements can form an ionic bond? (1791)

- A) Calcium and fluoride
- B) Nitrogen and chlorine
- C) Sulfur and hydrogen
- D) Carbon and oxygen
- E) Chlorine and bromine

24. If one mole of magnesium and one mole of oxygen gas completely react to give magnesium, how much product is formed? (383)

- A) 1 mole
- B) Half a mole
- C) 2 moles

25. Which molecule has the highest dissociation energy? (1348)

- A) Cl_2
- B) F_2
- C) N_2
- D) O_2
- E) Br_2

26. What do the standard conditions of an electrochemical cell refer to (63)

- A) 1 M, 1 atm, 298 K
- B) 1 atm, 273 K
- C) 1 atm, 298 K
- D) 1M, 298K

27. Which of these values can the quantum number n of an electron take (635)

- A) $1/2$
- B) -1
- C) 7
- D) h

28. Balance the following reaction and choose the correct sequence of coefficients for the products only. Acid phosphoric + $\text{Na}_2\text{O} \rightarrow \text{Na}_3\text{PO}_4 + \text{H}_2\text{O}$ (1321)

- A) 3, 3
- B) 2, 3
- C) 3, 2
- D) 2, 1
- E) 2, 2

29. It takes 48,000 coulombs to deposit $\frac{1}{2}$ mole of what metal? (2050)
- A) Aluminum
 - B) Silver
 - C) Iron
 - D) Magnesium
30. An aqueous solution is neither acidic nor basic when: (334)
- A) The concentration of H^+ ions is 10^{-7} mol/l
 - B) The concentration of H^+ ions is 10^7 molar
 - C) The concentration of H^+ ions is 10^{-14} mol/l
 - D) The pH is different from 7
 - E) The pH is equal to 10-14
31. the principle according to which an isolated system conserves its energy is (153)
- A) law 0 of thermodynamics
 - B) first law of thermodynamics
 - C) second law of thermodynamics
 - D) third law of thermodynamics
 - E) fourth law of thermodynamics
32. Indicate which of the following electronic configurations has the highest first ionization energy (1195)
- A) $1s^2 2s^2 2p^6 3s^2 3p^2$
 - B) $1s^2 2s^2 2p^2$
 - C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
 - D) $1s^2 2s^2 2p^5$
 - E) $1s^2 2s^2 2p^6 3s^2$
33. How many orbitals are filled in the atom with electron configuration $1s^2 2s^2 2p^6$? (1219)
- A) 2
 - B) 3
 - C) 4
 - D) 5
 - E) 6

34. which of the following molecules is polar (483)

- A) F₂
- B) HBr
- C) Br₂
- Q) H₂
- E) SF₆

35. What is one mole of gaseous fluorine equivalent to? (178)

- A) 38.0 Kg
- B) 38g
- C) 19g

36. How many molecules of products are generated by the combustion of 2 molecules of methane with oxygen?

(252)

- A) 2
- B) 4
- C) 6

37. Which of the following changes on the equilibrium: $4\text{NH}_3(\text{G}) + 3\text{O}_2(\text{G}) \rightarrow 2\text{N}_2(\text{G}) + 6\text{H}_2\text{O}(\text{G}) + \text{heat}$ do you experience an increase in NH₃ concentration? (1198)

- A) Addition of oxygen
- B) Lowering of pressure
- C) Increase in temperature
- D) Subtraction of nitrogen
- E) Lowering of T

38. In which of these compounds the oxygen does not reach the octet (582)

- A) X=O=X
- B) X=O: --->X
- C) X--> : O =X
- D) in all compounds reaches the octet

39. 1 mole of aluminum corresponds (189)

- A) 27 g of aluminum
- B) Avogadro's number of grams
- C) 56*6.022g
- D) 6.022×10^{22} molecules

40. How many metals are present in the following list: Cesium, Yttrium, Silicon, Argon (21)

- A) 3
- B) 1
- C) 2
- D) 4

41. Select exothermic process (268)

- A) Ice melting
- B) Boiling of water
- C) Condensation of water vapour
- D) Sublimation of ice
- E) None of these

42. What does 18 g of Be correspond to? (1752)

- A) 18 amu
- B) 9 protons
- C) 9 mol
- D) 2 mol
- E) none of the above

43. What the symbol || represents in the conventional notation of a galvanic cell? (206)

- A) a phase change
- B) the electrical connection
- C) salt bridge

44. 1 mole of water contains: (669)

- A) 32 g of oxygen
- B) Na hydrogen atoms
- C) Na molecules of oxygen
- D) 2*Na hydrogen atoms

45. Which of these bonds is the strongest? (477)

- A) N₂
- b) O₂
- C) HCl
- D) N₃

46. Which of these crystalline structures has the greatest packing factor? (1172)
- A) Face-centered cubic lattice
 - B) Body-centered cubic lattice
 - C) The structures are all equivalent
47. in a closed container P varies from 1 to 2 atm. If T was worth 1 degree, after that: (293)
- A) 1
 - B) 0.5
 - C) a value between 0.5 and 1
 - D) 2
48. Indicate the formula of bromic acid (1220)
- A) HBR
 - B) HBrO₃
 - C) HBrO₄
 - D) BrH₂
 - E) H₂Br₃
49. what is meant by alkyl group? (1308)
- A) an alkane deprived of a hydrogen atom
 - B) an alkene deprived of a hydrogen atom
 - C) an alcohol deprived of a hydrogen atom
50. The solidification of water is to be considered a process (265)
- A) Exothermic
 - B) Isoenthalpic
 - C) Isentropic
51. how many is the average molar mass of a gas mixture consisting of 50% by volume of CO (MM=28g/mol) and for the remainder from CO₂(MM=44g/mol) (479)
- A) 35.1 g/mol
 - B) 36 g/mol
 - C) 72 g/mol
 - D) 16 g/mol
 - E) none of the other answers are incorrect

52. In the following reaction what is the acid (according to Bronsted-Lowry) ? $\text{NH}_4^+ + \text{H}_2\text{O} \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+$

(1561)

- A) $\text{NH}_3(\text{aq})$
- B) H_3O^+
- C) does not exist
- D) NH_4^+
- E) H_2O

53. An aqueous solution of an electrolyte has, at the external pressure of 1 atmosphere, a point of boiling: (1213)

- A) $< 373\text{K}$
- B) $> 373\text{K}$
- C) $= 373\text{K}$
- D) $= 273\text{K}$
- E) $< 100\text{K}$

54. Carbonic acid is reacted with magnesium hydroxide. What do you get? (1678)

- A) MgCO_3
- B) MgCO_2
- C) MgCO
- D) MnCO_3
- E) MnCO_2

55. a catalyst is a substance which, when added to a reaction, modifies it (482)

- A) the enthalpy change
- B) the activation energy
- C) internal energy
- D) the variation of entropy
- E) the energy of the reactants

56. Balance the reaction hydrogen + nitrogen = ammonia. Indicates the stoichiometric coefficients of the reactants. (32)

- A) 3-1
- B) 2-3
- C) 4-5
- D) 6-7

57. Which of these is a reduction half-reaction? (647)

- A) $\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}$
- B) $\text{H}_2 + \text{e}^- \rightarrow 2\text{H}^+$
- C) $\text{H}_2 \rightarrow 2\text{H}^+ + \text{is}$

58. What happens when water boils? (665)

- A) Variation of vapor pressure
- B) Temperature is 100°C
- C) Enthalpy remains constant
- D) Entropy decreases
- E) None of the above correct

59. Which group does ethylene belong to? (1681)

- A) Alcohols
- B) aromatic compounds
- C) alkanes
- D) alkenes
- E) alkynes

60. Which of the following is not a property of alkali metals: (568)

- A) They form cations of $2+$ (x)
- B) Cerium is not an alkali metal
- C) They have 1 electron in the outer orbital

61. If you mix a solution containing 1g of HNO_3 (MM = 63) and one containing 1g of NaOH (mm = 40), the solution is: (1201)

- A) slightly acidic
- B) very acidic
- C) basic
- D) neutral
- E) very neutral

62. A real gas always kept above its critical temperature: (1210)

- A) It can be liquefied by compression
- B) It cannot be liquefied by compression
- C) It can be liquefied by expansion
- D) It can be liquefied with successive cycles of compression and expansion
- E) It may or may not be liquefied, depending on the conditions adopted

- 63.** Which of the indicated sequences correctly reports the compound formula, type of hybridization, geometry molecular and polarity (332)
- A) BeCl_2 /sp²/linear/non-polar
 - B) C_6H_6 /sp/hex/polar
 - C) CH_4 /sp³/tetrahedral/polar
 - D) NH_3 /sp²/pyramidal/polar
 - E) SO_2 /sp²/triangular/polar
- 64.** What is the volume of dry air with 0.84 moles of gaseous oxygen (368)
- A) $V = 100 \text{ L}$
 - B) V depends on the air pressure and temperature
 - C) $V > 200 \text{ L}$
 - D) $V < 200 \text{ L}$
 - E) none of the answers
- 65.** How much fluorine gas must react with hydrogen gas to give exactly one mole of HF? (577)
- A) 1 mole
 - B) 2 moles
 - C) 1 litre
 - D) None of the others
- 66.** The freezing point lowering of a solution containing a non-volatile solute is directly proportional to ... (1731)
- A) molality of the solute
 - B) mole fraction of the solvent
 - C) mole fraction of the solute
 - D) molarity of the solute
 - E) molality of the solvent
- 67.** Reduction is a phenomenon in which: (1137)
- A) An element reduces its oxidation number and acquires electrons
 - B) An element reduces its oxidation number and loses electrons
 - C) An element increases its oxidation number and acquires electrons
 - D) An element increases its number and loses electrons

68. Select the least soluble substance in Benzene (495)

- A) Pentane
- B) Decanol
- C) Octane
- Of water
- E) Hexane

69. A reaction with $\Delta H = -115.12$ is: (15)

- A) Exothermic, therefore it releases heat
- B) Endothermic, therefore it releases heat
- C) Exothermic, so it absorbs heat
- D) Endothermic, therefore it absorbs heat

70. Which orbital has $n=3$ $l=2$, $m=1$? (1702)

- A) Orbital s
- B) Orbital p
- C) Orbital d
- D) None
- E) Hybrid orbital

71. What is the Vantoff coefficient of sucrose? (388)

- A) 1
- B) 2
- C) infinity

72. A reaction at equilibrium has $K_p = \frac{[C]^2 [D]}{[A]^2 [B]}$, the reaction in question is: (1135)

- A) $2C + D \rightleftharpoons 2A + B$
- B) $2A + B \rightleftharpoons 2C + D$
- C) $A + B \rightleftharpoons C + D$
- D) $C + D \rightleftharpoons A + B$

73. At constant pressure, if you increase the volume by 50%, how does the temperature change (initially 100 K)?

(662)

- A) 100K
- B) 200K
- C) 50K
- D) 150K

74. Hydrogen Sulphide Formula (698)

- A) H_2SO_4
- B) H_2SO_3
- C) H_2S
- D) HS
- E) I don't remember

75. What is a property of alkali metals? (634)

- A) They do not form isoelectric ions with noble gases
- B) They form M^{++} ions
- C) They do not react with water
- D) Their first ionization energy decreases with increasing atomic weight

76. at 25° under what conditions is a solution acidic? (208)

- A) $[\text{H}_3\text{O}^+] > 10^{-7} \text{ mol/L}$
- B) $[\text{OH}^-] > 10^{-7} \text{ mol/L}$
- C) $\text{pH} > \text{pOH}$
- D) $[\text{H}_3\text{O}^+] > K_w$

77. How many hydrogen bonds can H_2S make (587)

- A) 4
- B) 3
- C) 2
- D) 1

78. Indicate from left to right the oxidation numbers of chlorine in the following ions: ClO_4^- , ClO_2^- , ClO^- , Cl^- (612) , chlorate anion,

- A) +5, +7, +3, +1
- B) +7, +5, +1, +3
- C) +1, +3, +7, +5
- D) +7, +5, +3, +1
- E) +3, +5, +7, +1

79. To increase the quantity of CO_2 in the container in which the following equilibrium is established : $\text{MgCO}_3 (\text{s}) \rightleftharpoons \text{MgO} (\text{s}) + \text{CO}_2 (\text{g})$ one must : (337)
- A) Add N_2 gas
 - B) Subtract MgCO_3 material
 - C) Increase the volume of the container
 - D) Add MgO
 - E) Increase pressure
80. Reduction in the voltaic cell occurs: (603)
- A) to the negative electrode
 - B) in the salt bridge
 - C) to the static electrode
 - D) to the positive electrode
 - E) there is not enough data to know it
81. what is molality given (597)
- A) n moles of the solute divided by the volume of the solution
 - B) grams of solute divided by weight of solvent
 - C) n of moles of the solute divided by the weight of the solvent
 - D) volume of solute divided by volume of solvent
82. Which of these compounds is not polar (641)
- A) CO_2
 - B) HCl
 - C) HF
 - D) HI
 - E) H_2O
83. There are two containers of 100 ml containing one Nitrogen and the other Oxygen at pressure conditions and standard temperature. They have: (1356)
- A) same number of molecules
 - B) same average speed of the molecules
 - C) none of the above
84. Given a reaction $\text{A} + 2\text{B} = 3\text{C}$ and given a second reaction $\frac{1}{3}\text{A} + \frac{2}{3}\text{B} = \text{C}$ what is the ratio between K_{c1} and K_{c2} ? (296)
- A) $K_{c1} = \text{rad}3(K_{c2})$
 - B) $K_{c1} = (K_{c2})^3$
 - C) $K_{c1} = K_{c2}$

85. What is the amount of CaCO_3 that exactly 40g of calcium contains? (241)

- A) 2 moles
- B) 100g
- C) 23 dm^3
- D) 64g
- E) None of the above

86. If a solution is basic: (269)

- A) $[\text{H}^+] > [\text{OH}^-]$
- B) $K_w < 10^{-7}$
- C) $K_w = 10^{-7}$
- D) $K_w > 10^{-7}$
- E) None of these

87. In a container there is a hydrocarbon and some oxygen and it is reacted, obtaining steam and CO_2 . As is changed the density of the mixture? (693)

- A) augmented
- B) diminished
- C) depends on the gas
- D) has not changed
- E) no previous

88. Which of cis-2-butene and trans-2-butene has a non-zero dipole moment (1751)

- A) Cis
- B) Trans
- C) None
- D) Both
- E) There is no isomerism

89. What is the chemical formula of butanoic acid? (598)

- A) $\text{C}_4\text{H}_8\text{O}_2$
- B) C_4O_2
- C) $\text{C}_4\text{H}_6\text{O}_2$

90. The secondary quantum number can be associated with which of the following characteristics of motion of the electron? (1521)
- A) The energy of the orbital in the absence of a magnetic field
 - B) The number of electrons
 - C) None of the other features
 - D) The spatial orientation of the orbital
 - E) The shape of the orbital
91. C_nH_{2n-2} are.. (386)
- A) alkanes
 - B) alkenes
 - C) alkynes
 - D) hydroxides
 - E) none of the above
92. What is the complete reaction of electrolysis of water (558)
- A) $H_2O = H_3O^+ + OH^-$
 - B) $2H^+ + 2e^- = H_2 (g)$
 - C) $2H_2O = 2H_2 + O_2$
93. How do I lower the freezing point of a compound saturated with KCl? (1243)
- A) I add KCl
 - B) I increase the pressure that weighs on the solution
 - C) I reduce the pressure that weighs on the solution
 - D) I add water
94. Aromatic compound (1553)
- A) Cyclohexane
 - B) Toluene
 - C) Exene
95. Which element has the smallest atomic radius (31)
- A) Cs
 - B) H
 - C) That
 - D) K
 - E) No

96. the water molecule has geometry (427)

- A) Linear
- B) Planar trigolane
- C) Tetragonal
- D) Angular

97. How many structural isomers does dibromobenzene have? (6)

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5

98. A reaction is always spontaneous if: (9)

- A) $\Delta S < 0$
- B) $\Delta H > 0$
- C) $\Delta H = T\Delta S$
- D) $\Delta H < T\Delta S$
- E) $\Delta H > T\Delta S$

99. Diamond and graphite are _____ carbon (1851)

- A) Isomers
- B) Leagues
- C) Isotopes
- D) Allotropes
- And more

100. What does NOT affect reaction speed? (364)

- A) Presence of a catalyst
- B) Surface of solid reactants
- C) All these factors influence the speed
- D) Temperature

101. Select what you cannot find in the Daniell Stack (1322)

- A) Cu^+ ions
- B) Zinc ions
- C) Salt bridge
- D) Spontaneous reaction
- E) SO_4^{2-} ions

102. Under which of the following conditions can the reaction not be spontaneous? (95)

- A) $\Delta H < 0$ $\Delta S < 0$
- B) $\Delta G < 0$
- C) $\Delta H < 0$ $\Delta S > 0$
- D) $\Delta H > 0$ $\Delta S > 0$
- E) $\Delta H > 0$ $\Delta S < 0$

103. How do you convert 67 atm into millimeters of Mercury? (1831)

- A) 67/760
- B) 67+760
- C) 67*670
- D) 67/670
- E) None of the above

104. $\text{CH}_3\text{-CH}_2\text{-NH}_2$ (166)

- A) It is a strong foundation
- B) It is a weak base
- C) It is a strong acid
- D) It is a weak acid
- E) It is an Arrhenius basis

105. during the evaporation of water what certainly happens? (1161)

- A) the entropy remains unchanged
- B) the vapor pressure increases
- C) the water is at 100°C
- D) increases entropy
- E) none of the answers is correct

106. How many moles of water can I get From one mole of O_2 is one mole of H_2 (312)

- A) 0 moles
- B) 0.5 moles
- C) 1 mole
- D) 2 moles

107. The quantum number principle n what value can it assume? (200)

- A) h
- B) +7
- C) -1
- D) +1/2

108. To predict the degree of polarity of a molecule it is necessary to know: (345)

- A) The electronegativity of its atoms and its geometry
- B) The electronegativity of its atoms
- C) The oxidation number of its atoms
- D) Its geometry
- E) Its structural formula

109. Given a graph of an exothermic reaction (y axis -> Energy), how does the graph vary in the presence of a catalyst? (87)

- A) Lower the maximum point
- B) raise the maximum point
- C) raise the coordinate of the reactants
- D) raise the coordinate of the products
- E) lower the coordinate of the reactants

110. In an isothermal transformation, the product p by V (182)

- A) It is constant in ideal gases
- B) It is constant both in real gases and in ideal gases
- C) Does not depend on R
- D) None of the answers are complete

111. Select the best percent composition by volume of dry air: (2467)

- A) it depends on the pressure
- B) 21% Argon, 78% Oxygen, 1% Nitrogen
- C) depends on the temperature
- D) 21% Nitrogen, 78% Oxygen, 1% Argon
- E) 21% Oxygen, 78% Nitrogen, 1% Argon

112. What principle of thermodynamics highlights the irreversibility of chemical reactions? (313)

- A) 1 principle of thermodynamics
- B) 2 principle of thermodynamics
- C) 0 principle of thermodynamics
- D) 3 principle of thermodynamics
- E) Le Chatelier's principle

113. Formula of Phenol (1325)

- A) C_6H_5OH
- B) C_6H_6
- C) CH_3OH

114. How many phases does the following reaction have: $\text{HF(g)} + \text{H}_2\text{O(aq)} \rightleftharpoons \text{H}_3\text{O}^+(\text{g}) + \text{F}^-(\text{g})$

(1732)

- A) 2
- B) 3
- C) 1
- D) 5
- E) 0

115. What happens when current is passed through an aqueous solution of KOH? (695)

- A) electrolysis of water
- B) precipitates K
- C) K_2O precipitates
- D) nothing happens
- E) none of the above

116. The ΔH of an exothermic reaction is (250)

- A) >0
- B) <0
- C) $\Delta H > T\Delta S$
- D) $=0$
- E) $\Delta H < T\Delta S$

117. In a pressure-volume diagram at constant temperature the trend of the curves is (621)

- A) Exponential
- B) Logarithmic
- C) Hyperbolic
- D) Quadratic
- E) Linear

118. You have two moles of hydrogen peroxide, N_A being Avogadro's number. In the molecule they are present (1144)

- A) $4/N_A$ atoms of oxygen and $4/N_A$ of Hydrogen
- B) $2/N_A$ atoms of oxygen and $2/N_A$ of Hydrogen
- C) 2 moles of atoms
- D) $4 \cdot N_A$ oxygen atoms and $4 \cdot N_A$ hydrogen atoms
- E) $2 \cdot N_A$ oxygen atoms and $2 \cdot N_A$ hydrogen atoms

119. a mole of iron contains (434)

- A) Na iron atoms
- B) 111g
- C) Na molecules of iron
- D) 56 Kg
- E) Na iron particles

120. According to one theory, a chemical species is defined as a basis which.. (1647)

- A) accepts an electron
- B) donate an electron
- C) reacts with water
- D) accepts a proton
- E) donates a proton

121. $\text{Fe}(\text{OH})_3 + \text{H}_2\text{SO}_4 \rightarrow$ (161)

- A) FeSO_4
- B) FeSO_3
- C) $\text{Fe}_2(\text{SO}_4)_3$
- D) FeS
- E) $\text{Fe}(\text{SO}_3)_3$

122. What type of structure does ammonia have? (431)

- A) trigonal planar
- B) octahedral
- C) tetragonal
- D) trigonal pyramid

123. In a graph of pressure vs temperature for an isochore, the graph of temperature as a function of pressure has the following trend: (1539)

- A) quadratic
- B) linear
- C) logarithmic
- D) exponential
- E) constant

124. Which of these does not respect the octet rule? (1768)

- A) NH_3
- B) PCl_5
- C) H_2O
- D) S_2
- E) NH_4^+

125. Which of the following statements about the diamond is false: (1203)

- A) is a covalent solid of tetrahedral symmetry
- B) Carbon sp^3 hybridizes and connects to 4 other carbon atoms
- C) fails to conduct electricity
- D) is prepared by synthesis from graphite
- E) carbon is not hybridized

126. The volume fraction of oxygen in a gas mixture is 0.2. What is the volume of mixture that contains 40L of oxygen? (1520)

- A) It depends on the air pressure and temperature
- B) 16.6 dm^3
- C) 164 L
- D) 200 L
- E) 8 L

127. What maximum value of m can an element of the sixth period in its ground state assume? (292)

- A) -1
- B) 1
- C) 2
- D) 3

128. Esters are compounds resulting from the reaction between an alcohol and a: (1165)

- A) alkane
- B) ketone
- C) carboxylic acid
- D) ether
- E) heel

129. What principle states that an isolated system retains all its energy (1531)

- A) Maxwell-Boltzmann
- B) third law of thermodynamics
- C) zeroth law of thermodynamics
- D) second law of thermodynamics
- E) first law of thermodynamics

130. How many isomers of C_4H_{10} are there? (692)

- A) 1
- B) 2
- C) 4
- D) 6
- E) This formula does not exist

131. Which of these factors never influences the total heat developed by a chemical reaction? (338)

- A) The rate of the reaction
- B) The reaction temperature
- C) The crystalline modification of reactants and products if solid
- D) The pressure and volume conditions in which the reaction takes place
- E) The physical state of reactants and products

132. An ideal gas occupies a volume of 4.5 L at 300K. Considering P constant, at which T the gas occupies a volume of 9.0 L? (109)

- A) 150K
- B) 150°C
- C) 600K
- D) 373K
- E) 300K

133. One substance is reducing with respect to another if: (333)

- A) It has less tendency to pass into the ionic form
- B) None of the other answers
- C) It has a greater tendency to lose electrons
- D) It has a greater tendency to acquire electrons
- E) It shrinks more easily

134. What is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$? (602)

- A) propanol
- B) butane
- C) 1-butanol
- D) butanoic acid
- E) butane alcohol

135. Which of these units of measurement does not refer to pressure? (168)

- A) Torr
- B) Pascal
- C) Debye
- D) mmHg
- E) Bar

136. The theory of ____ says that "acids in water release H^+ ions" (691)

- A) Arrhenius
- B) Bronsted-Lowry
- c) London
- Q) Lewis

137. Which of these items is not contained in the Daniell pile (1143)

- A) $(\text{SO}_4)^{2-}$
- B) Cu^+
- C) Zn^{2+}

138. To which category does C_6H_{12} belong? (362)

- A) There is no single answer
- B) Alkenes
- C) Aromatics
- D) Cycloalkanes

139. select conventionally used symbol for entropy (497)

- A) St
- B) H
- C) sigma Q
- D) E
- E) omegas

140. How many neutrons are in deuterium? (1318)

- A) 2
- B) 3
- C) 1
- D) It depends on the isotope

141. How many isomers are there of dichlorobenzene? (64)

- A) 1
- B) 3
- C) 5
- D) 6

142. How much oxygen gas does it take to oxidize 46g of sodium? (1171)

- A) 1 mole
- B) 2 moles
- C) 0.5 moles
- D) 32 grams
- E) 23 grams

143. The mean molar mass of air depends on: (1767)

- A) Both by relative humidity and by temperature
- B) Only from relative humidity and not from temperature
- C) Only from the temperature and not from the relative humidity
- D) It cannot be calculated
- E) Neither by relative humidity nor by temperature

144. What is the volume of one kg of liquid ice at 4°C? (317)

- A) you need to know the density
- B) 0.9l
- C) 1000 cc
- D) 1000 cl
- E) there is no liquid ice cube at that temperature

145. Which of these cannot be obtained by oxidation of ethanol? (319)

- A) acetic acid
- B) ethanoic acid
- C) CH_3COOH
- D) all answers are valid

146. Knowing that the molecular mass of water is equal to 18 amu, in a liter of water are contained: (1795)

- A) 22.4 mol of water
- B) 55.5 mol of water
- C) 111 mol of water
- D) 55.5 moles of copper

147. Acetic acid has pH=6. What is the pH of the corresponding aqueous solution? (622)

- A) I have no elements to answer
- B) 10^{-6} M
- C) 10^{-8} M
- D) 10^6 M
- E) 10^8 M

148. How can the equilibrium constant vary (594)

- A) Change concentrations
- B) Temperature
- C) Addition of catalyst
- D) None of these

149. What is the IUPAC name of CO? (1147)

- A) Carbon dioxide
- B) Carbon peroxide
- C) Carbon monoxide

150. what does the binding energy NOT depend on? (1242)

- A) binding order
- B) atomic radius
- C) it depends on everyone
- D) bond distance
- E) electronegativity

151. If you dilute a solution (134)

- A) The moles of solute remain unchanged
- B) Moles of solute increase
- C) Moles of solute decrease
- D) Moles of solvent decrease
- E) The ratio of moles solute to mole solvent increases

152. An aqueous solution of hydrochloric acid has $\text{pH} = 4$. If I dilute it with an equal volume of water, the pH :

(1197)

- A) remains unchanged
- B) decreases slightly
- C) increases slightly
- D) double (8)
- E) decreases by one unit

153. Which compound reacts with H_2O to give nitric acid? (632)

- A) N_2O_5
- B) N_2O_3
- C) H_2
- D) N_2
- E) NH_4^+

154. Given two half reactions with relative enthalpies, how is the enthalpy of the total reaction calculated? (1341)

- A) adding the enthalpies
- B) dividing the enthalpies
- C) by subtracting the enthalpies
- D) multiplying the enthalpies

155. How many isomers does C_4H_{10} have? (1342)

- A) 0
- B) 3
- C) 2
- D) 4
- E) C_4H_{10} does not exist

156. What does the partial pressure of a gas in a gaseous mixture of ideal gases depend on? (403)

- A) Volume
- B) Temperatures
- C) Mole fraction
- D) Volume fraction
- E) Type of gas

157. How many singular covalent bonds must an oxygen atom form to reach the octet? (494)

- a) 5
- B) 2
- C) 3
- D) 4

158. what is this compound called? $\text{H}_2\text{-C=O}$ (290)

- A) methane
- B) methanal
- C) dihydrogen aldehyde

159. Li (with $E^0 = -3\text{V}$) and Be ($E^0 = -2\text{V}$) are introduced into a voltaic cell. What is the difference of head tension? (1167)

- A) 1 V
- B) -1 V
- C) +5 V

160. The density of a gas is (1682)

- A) Directly proportional to the temperature
- B) None of the above
- C) Inversely proportional to the molar mass
- D) Inversely proportional to the temperature

161. Knowing that the pH of a solution is 6 then the concentration of the acetic acid is : (410)

- A) $10^{(-6)}\text{M}$
- B) $10^{(6)}\text{M}$
- C) $10^{(-12)}\text{M}$
- D) There is not enough data

162. Which of the following formulas can represent ethylbenzene? (2096)

- A) C_7H_{10}
- B) C_7H_8
- C) C_8H_{12}
- D) C_8H_{10}
- E) C_8H_8

163. Mass number is (1141)

- A) Sum neutrons and protons in a nucleus
- B) Sum protons electrons and neutrons in a nucleus
- C) Number of protons
- D) Number of neutrons
- E) Difference between protons and neutrons

164. Which of the following systems is NOT heterogeneous? (704)

- A) Foam
- B) Suspension
- C) I smoke
- D) Emulsion
- E) solution

165. What is the correct definition of acid salt? (1168)

- A) Hydracid with addition of water
- B) Salt that derives from the non-total dissociation of the hydrogen atoms in the acid molecules
- C) Molecules containing only H, O and no metals

166. X-rays are (289)

- A) high frequency electromagnetic waves
- B) low frequency electromagnetic waves
- C) electrons
- D) protons
- E) alpha particles

167. In which case is a solution spontaneous only at "high" temperatures? (10)

- A) $\Delta H > 0$ and $\Delta S > 0$
- B) $\Delta H < 0$ and $\Delta S > 0$
- C) $\Delta H < 0$ and $\Delta S < 0$
- D) $\Delta H > 0$ and $\Delta S < 0$

168. A gas under normal conditions is isothermally compressed until it reaches a pressure equal to 3 times that atmospheric. What happens to the volume of the gas? (1554)

- A) Decreases by 1/3
- B) It becomes equal to 1/3 of the initial one
- C) It increases slightly
- D) Triples
- E) Remains unchanged

169. What is an alkyl group (580)

- A) An alkane with one less H
- B) An alkene with one less H
- C) An alkyne with one less H

170. Which of the following is the structural formula of glycerol (also called glycerin) (636)

- A) $C_3H_8O_3$
- B) $C_3H_5O_3$
- C) C_3H_8

171. Who formulated the law of conservation of mass? (1581)

- A) Democritus
- B) Lavoisier
- C) Daltons
- Q) Lewis

172. The anode of the Daniell cell is formed by: (247)

- A) a sheet of zinc immersed in a solution with zinc ions
- B) a zinc foil immersed in a solution with copper ions
- C) a copper foil immersed in a solution of copper ions
- D) a copper foil immersed in a solution of zinc ions

173. Which of the following cannot be obtained by oxidation of 2-propanol? (696)

- A) CO_2
- B) propanone
- C) propanoic acid
- D) (another ketone)
- E) all of the above

174. which of the following processes is used to remove ions from water via a membrane semipermeable? (204)

- A) reverse osmosis
- B) distillation
- C) precipitation
- D) filtration

- 175.** A picture is given with a molecule with different bond types, each bond is numbered. he wonders to find the strongest bond (681)
- A) double
 - B) single
 - C) triple
- 176.** Carbon 14, what is 14? (426)
- A) Atomic weight
 - B) Atomic mass
 - C) Number of moles per litre
- 177.** Which group contains only hydracids? (384)
- A) HI, HF, HBr, H₂S
 - B) HF, HBr, HCN, H₂SO₄
 - C) HI, HN₃, H₂SO₃
- 178.** In the electrolytic cell the anode represents the pole (651)
- A) positive
 - B) negative
 - C) where the reduction takes place
- 179.** I have a gas at 100K and the volume increases by 50%. What is the temperature at constant pressure? (404)
- A) It is not possible to answer
 - B) 100k
 - C) 50K
 - D) 150K
 - E) 200k
- 180.** What was the first particle to be discovered, studied and described? (1779)
- A) Proton
 - B) Neutron
 - C) Electron
 - D) Positron
 - E) Neutrino
- 181.** The catalyst in a reaction: (694)
- A) decreases the activation energy
 - B) increases entropy
 - C) enthalpy decreases

182. what these H_2SO_4 / Na_2SO_3 compounds correspond to (205)

- A) sulfuric acid / sodium sulfite
- B) hydrogen sulphide / sodium sulfate
- C) sulfuric acid / sodium sulfate
- D) sulfurous acid / sodium sulfite
- E) sulfurous acid / sodium sulfate

183. An aqueous solution of ammonia has a $\text{pH}=12$. If I dilute it with an equal volume of water the

pH : (336)

- A) is halved (it becomes equal to $12/2$ i.e. 6)
- B) Increase by one unit
- C) remains unchanged
- D) decreases slightly
- E) increases slightly

184. Reagent Coefficients $\text{C}_6\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (1541)

- A) 2;8
- B) 1;4
- C) 2;15
- D) 1;13
- E) 1;7

185. To form 34g of H_2O_2 , what is the minimum amount of oxygen required? (107)

- A) 32g
- B) 2 molecules
- C) 1 molecule
- D) 2 moles of diatomic gas
- E) 16g

186. Given 500mL of nitrogen at a pressure of 0.75atm, what will be the volume at a pressure of 1.50atm, assuming the temperature remains constant? (1317)

- A) 125mL
- B) 1000mL
- C) 2000 mL
- D) 250mL
- E) 500 mL

187. Identify the correct pair of names for the following compounds: H_2SO_3 and Na_2SO_4 (1738)

- A) sulfuric acid / sodium sulfite
- B) sulfuric acid / sodium sulfate
- C) sulfurous acid / sodium sulphate
- D) hydrogen sulphide / sodium sulfate
- E) sulfurous acid / sodium sulfite

188. Let an atom be $Z=9$, $A=19$, write in the order the number of electrons, neutrons, unpaired electrons (96)

- A) 9, 10, 1
- B) 1, 9, 10
- C) 19, 10, 9
- D) 9, 19, 1
- E) 9, 10, 0

189. 14 moles of CaO contain: (1170)

- A) $(14 \cdot 6.022)$ Ca molecules and $(14 \cdot 6.022)$ O molecules
- B) 14g of Ca and 14g of O
- C) 14Na atoms of Ca and 14Na atoms of O
- D) $(14/6.022)$ Ca atoms and $(14/6.022)$ O atoms
- E) 14/O molecules of Ca and 14/O molecules of O

190. In the CO_2 molecule are involved in total: (339)

- A) a pi bond and two sigmas
- B) two pi bonds and a delocalized sigma
- C) two pi bonds + two sigmas
- D) a delocalized pi bond and two sigmas

191. Which of the following cannot have the electron configuration $1s^2 2s^2 2p^6$? (630)

- A) Transition metals in their ground state
- B) Alkali metal cations
- C) Noble gases

192. A gas occupies 500mL at 0.75 atm, how much it occupies at 1.5 atm at constant T (76)

- A) 200 mL
- B) 250ml
- C) 1000ml
- D) 2000ml
- E) 2dm^3

193. Which of the following substances is not typically reducing? (485)

- A) nitric acid
- B) magnesium
- C) iron
- D) hydrogen
- E) carbon

194. An ester can be obtained from the reaction between an alcohol and (1584)

- A) Aldehyde
- B) Ketone
- C) Carboxylic acid
- D) Amine

195. taking an atom of Ca, how much does it weigh? (1786)

- A) 40.08g
- B) 40.08 auma
- C) 40.00 au
- D) It depends on which isotope we are considering

196. What is the definition of acid salt? (91)

- A) A salt in which the acidic hydrogens have not been completely replaced
- B) a salt that derives from a strong salt
- C) a salt that derives from a weak salt

197. Arrange the following compounds in decreasing order of bond strength (11)

- A) HF, HCl, HBr, HI
- B) HBr, HF, HI, HCl
- C) HF, HBr, HI, HCl
- D) HCl, HF, HBr, HI

198. Traditional nomenclature of: NaOH (1221)

- A) Oxidized sodium
- B) Sodium hydroxide
- C) Sodium oxide
- D) Nitrogen hydroxide
- E) Oxygenated sodium hydride

199. What is the strongest bond? (267)

- A) DC
- B) CH
- C) C=C
- D) C $\ddot{y}$ C
- ECHO

200. Which of these molecules has the greatest binding strength (1336)

- A) nitrogen gas
- B) gaseous oxygen
- C) nitric acid
- Of water

201. IUPAC nomenclature of the lowest molar weight ketone (220)

- A) Propanone
- B) Methanone
- C) Ethanone
- D) Acetone
- E) Butanone

202. In a solution at 25°C to make it basic... (105)

- A) pH < pOH
- B) pOH > 10⁷
- C) none of the above

203. choose the right raw formula of benzene (273)

- A) C₆H₆
- B) C₆H₁₂O₆
- C) C₄H₆

204. Given a pressure-temperature graph, at constant volume, when does the graph touch the abscissa axis?

(377)

- A) 0 K
- B) 0°C
- C) 273 K
- D) 273°C
- E) At no point

205. Which of this element has the highest first ionization energy? (1158)

- A) O
- B) F
- C) H
- D) Ne
- E) Kr

206. One mole of fluorine corresponds to: (235)

- A) Avogadro's number moles of fluorine atoms
- B) 38 kg of fluorine
- C) Avogadro's number moles of fluorine molecules
- D) 19 g of fluorine
- E) 19×10^{23} fluorine atoms

207. Select endothermic process: (353)

- A) Solidification of water
- B) Condensation of water vapour
- C) Solvation of zinc chloride
- D) Metal burning
- E) Melting of solid zinc

208. Indicate the correct sequence for the boiling temperature at the pressure of 1 atmosphere for the series of HX compounds (X = F, Cl, Br, I) (343)

- A) $\text{HI} < \text{HBr} < \text{HF} < \text{HCl}$
- B) $\text{HF} < \text{HI} < \text{HBr} < \text{HCl}$
- C) $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$
- D) $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$
- E) $\text{HCl} < \text{HI} < \text{HBr} < \text{HF}$

209. How many grams do 2 moles of water weigh? (297)

- A) 2
- B) 8
- C) 18
- D) 24
- E) 36

210. select the correct definition of an endothermic reaction (491)

- A) reaction that takes place at low pressure
- B) reaction that takes place at high pressure
- C) heat-absorbing reaction
- D) reaction that gives off heat
- E) I don't remember the last proposal anymore (he was talking about activation energy)

211. What kind of transformation is freezing water? (1757)

- A) isenthalpic
- B) isentropic
- C) exothermic
- D) None of the above

212. Two different organic substances having the same crude formula possess: (1562)

- A) The same boiling point
- B) The same density
- C) The same melting point
- D) The same viscosity
- E) The same percentage composition

213. in a tank there are 8 Moles of hydrogen plus 6 moles of Oxygen they form water, at the end of the reaction when the reagents are completely used up what is left in the tank? (687)

- A) Water
- B) Water and hydrogen
- C) Hydrogen and oxygen water
- D) Water and oxygen

214. A solution has pH = 6. The concentration of H_3O^+ ions is: (82)

- A) - log6
- B) -log(-6)
- C) 10^6
- D) 10^{-6}
- E) 6^{10}

215. Which of these molecules is linear? (1196)

- A) H₂S
- B) CO₂
- C) CH₃-CH₃
- D) SO₂
- E) CH₄

216. What is the name of the compound HClO_4 ? (1549)

- A) Hypochloric acid
- B) Hydrochloric acid
- C) None of the answers are correct
- D) Chloric acid
- E) Chlorous acid

217. The electron configuration: [Ar]3d⁶4s⁰ represents the ion: (1209)

- A) Mn²⁺
- B) Ni²⁺
- C) Fe³⁺
- D) Co³⁺
- E) Cu²⁺

218. $AV = \text{cost}$. If I increase T in a gas, which of these properties does it change? (143)

- A) Pressure
- B) The relationship between pressure and temperature
- C) At least two of the answers

219. What is the maximum number that the secondary quantum number l can assume in the current table periodic (578)

- A) Infinity
- B) 3
- C) 4
- D) 6

220. Which of the following oxides reacts with water to form a hydroxide? (484)

- A) SO₂
- B) CO
- C) CO₂
- D) N₂O₅
- E) none of the other answers are correct

- 221.** Calculate the mean molar mass of a mixture composed of 10% by volume of O₂ and 90% by volume of O₂ by H₂ (1782)
- A) 5 g/mol
 - B) 7 g/mol
 - C) 3 g/mol
 - D) None are correct
 - E) 9 g/mol
- 222.** How many grams of S are there in 2 moles of H₂SO₄ (227)
- A) 640g
 - B) 32g
 - C) 0.064 Kg
 - D) 1 mol
- 223.** Select the aqueous solution with the highest osmotic pressure (1517)
- A) 0.1M BaCl₂
 - B) 0.1M NaCl
 - C) 0.1M CuSO₄
 - D) 0.1M C₆H₁₂O₆
 - E) 0.1M HNO₃
- 224.** Balance the following reaction and select the stoichiometric coefficients of the reactants in the correct sequence copper sulphate(II) + Fe(NO₃)₃ _____ iron sulphate(III) + Cu(NO₃)₂ (487)
- A) 2.3
 - B) 3.3
 - C) 1.1
 - D) 3.2
 - E) 2.2
- 225.** Arrange the following compounds in ascending order by oxidation number O₂, H₂O₂, Na₂O, F₂O, CO₂ (151)
- A) F₂O, Na₂O, CO₂, O₂, H₂O₂
 - B) CO₂, Na₂O, H₂O₂, O₂, F₂O
 - C) CO₂, Na₂O, F₂O, H₂O₂, O₂
 - D) F₂O, O₂, H₂O₂, Na₂O, CO₂
 - E) none of the above

226. what cannot be achieved by oxidizing 2-propanol (152)

- A) propanoic acid
- B) propanone
- C) methanol
- D) dimethyl ketone

227. Which of the following molecules is not polar? (1177)

- A) Br₂
- B) HCl
- C) H₂O

228. Aqueous solution with higher boiling temperature (1189)

- A) HNO₃
- B) SO₄
- C) C₈H₂₂O₁₂
- D) CH₄

229. To make a reaction be spontaneous at low temperatures (162)

- A) $\Delta G > 0$
- B) $\Delta H < 0$ $\Delta S > 0$
- C) $\Delta H < 0$ $\Delta S < 0$
- D) $\Delta H > 0$ $\Delta S > 0$
- E) $\Delta H > 0$ $\Delta S < 0$

230. The maximum value of n is (1217)

- a) 5
- B) 3
- C) 7

231. How many of these elements are metals: Itrium, Cesium, Radon, Silicon (1766)

- A) 0
- B) 2
- C) 4
- D) 3
- E) 1

232. What is a secondary alcohol oxidized to? (1353)

- A) Aldehyde
- B) Carboxylic acid
- C) Ketone
- D) Ester
- E) None of the above

233. What is the maximum value that the secondary quantum number l can assume in the current periodic table

(1734)

- a) 7
- B) 6
- C) infinity
- D) 1
- E) 3

234. What is obtained from $\text{HNO}_2 + \text{KOH}$? (1215)

- A) $\text{KNO}_2 + \text{H}_2\text{O}$
- B) KNO
- C) $\text{KNO}_5 + \text{H}_2\text{O}$

235. How many resonances of limit structure are there in nitric acid? (90)

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5

236. what is the process that causes the movement of solvent from the less concentrated solution to the more focused? (439)

- A) distillation
- B) filtration
- C) osmosis
- D) titration
- E) reverse osmosis

237. Indicate the molecular geometry of methane. (1548)

- A) Trigonal Planar
- B) Linear
- C) Tetrahedral
- D) Trigonal pyramidal
- E) Octahedral

238. Why is the standard reduction potential of hydrogen (Standard Hydrogen Electrode - SHE) zero? (88)

- A) It is by convention
- B) because it oxidizes easily
- C) Does not generate voltage in any case

239. Which of the following statements is correct at the equilibrium of a solution? (1244)

- A) $K=0$
- B) the forward and reverse speeds coincide
- C) the reaction stops

240. A solution can never be (417)

- A) polyphasic
- B) unsaturated
- C) saturated
- D) metastable
- E) solid

241. Electron configuration of nitrogen (71)

- TO)
- B)
- C)
- D)

AND)

242. select a correct definition of second ionization energy for chlorine (486)

- A) the energy required to form two $\text{Cl}^{(-)}$ ions from two Cl atoms
- B) the energy released by a Cl atom to form $\text{Cl}^{(2-)}$
- C) the energy required to remove and carry an electron infinitely away from $\text{Cl}^{(+)}$
- D) the energy required to add two electrons to Cl
- E) the energy required to remove and carry an electron from $\text{Cl}^{(-)}$ to an infinite distance

243. Indicate the electronic configuration of the lithium ion: (79)

- A) [He]1s²
- B) [He]2s⁰
- C) [He]2s² 2p⁵
- D) [He]2s²
- E) [He]2s¹

244. One mole of a certain substance weighs 50 g; how many grams of this substance will I have at a pressure of 1 atmosphere if I have a volume of 2 liters, at a temperature of 100K? (604)

- A) 25/0.0821 g
- B) 100/3.814 g
- C) 100/(3.814*373) g
- D) 1/0.0821g

245. In which class of oxygenated compounds the C=O group is not present: (1564)

- A) aldehydes
- B) alcohols
- C) esters
- D) carboxylic acids
- E) ketones

246. How many molecules does 1.5 moles of NO have? (Na = avogadro's number) (1750)

- A) 6.022×10^{23}
- B) 1.5×6.022
- C) $1.5 \times 6.022 \times 10^{23}$
- D) Na
- E) $1.5/\text{Na}$

247. For an equilibrium reaction, a K value of 10^{-4} indicates that the equilibrium... (1794)

- A) favors the reactants
- B) favors the products
- C) depends on the temperature
- D) does not favor anyone

248. for a solution to be spontaneous at a low temperature there must be the following condition (1152)

- A) $\Delta G > 0$
- B) $\Delta H > 0$ $\Delta S > 0$
- C) $\Delta H > 0$ $\Delta S < 0$
- D) $\Delta H < 0$ $\Delta S < 0$
- E) $\Delta H < 0$ $\Delta S > 0$

249. What is the minimum number of carbon atoms to have a structural isomer? (492)

- A) 3
- B) 2
- C) 1
- D) 4
- E) 5

250. two sealed gas cylinders have volume $V=1$. One is filled with hydrogen, the other with nitrogen. What can you to say? (564)

- A) have the same number of particles
- B) have equal density
- C) have different densities

251. The electromotive force of a voltaic cell at chemical equilibrium is: (8)

- A) It depends on the reaction
- B) < 0
- C) Infinity
- D) 1

252. Which substance contains a methyl group: (1351)

- A) Cyclohexane
- B) Trinitrobenzene
- C) Acetylene
- D) Methanoic acid
- E) Toluene

253. Which of the following compounds has only covalent bonds? (108)

- A) SO_3
- B) PbS
- C) NaCl
- D) BaSO_4
- E) HCl

254. Consider the following equilibrium: $2\text{HI (g)} \rightarrow \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$. What happens if you increase the pressure partial of HI ? (352)

- A) The equilibrium shifts to the right
- B) Nothing
- C) Equilibrium K increases
- D) The balance shifts to the left
- E) Increases reaction speed

255. How many liters of water must be added to 500 ml of a 2M solution to change to 1M (1156)

- A) 0.25
- B) 0.5
- C) 1
- D) 2
- E) 4

256. I have 0.8 moles of solute dissolved in 1.2 moles of solvent. What is the mole fraction of the solute (279)

- A) 0.4
- B) 0.8
- C) $0.4/(1.2+0.8)$

257. What happens when you dilute a solution? (697)

- A) the number of moles of the solvent decreases
- B) the ratio of moles of solute to moles of solvent increases
- C) the number of moles of solute particles remains constant
- D) the product between moles of solute and solvent remains constant

258. Which of the following substances is the most soluble in water? (98)

- A) Pentane
- B) Ethanol
- C) Cyclohexane
- D) Decanol
- E) Etino

259. Calculate the osmotic pressure of a 1M solution where $V=2\text{ L}$ and temperature T (1159)

- A) $0.5 \times R \times T$
- B) $1 \times R \times T$
- C) $2 \times R \times T$

260. 2 molecules of Hydrogen and 2 molecules of Oxygen are in a reactor. How many molecules are there at the end of the water formation reaction in the reactor? (675)

- A) 1
- B) 2
- C) 3
- D) 4

261. Name of CuO (607)

- A) copper monoxide
- B) cuprous oxide
- C) cupric oxide
- D) copper(I) oxide

262. 44.8 L is the volume occupied by: (335)

- A) Two moles of any ideal gas under any condition
- B) Two moles of any gas at 25°C and 1.00 atm
- C) None of the other answers
- D) Two moles of any ideal gas at 273 K and 760 mmHg
- E) Two moles of any ideal gas at 100C and 1.00 atm

263. What characteristics do elements of the same group have in common? (20)

- A) Same number of electrons in the outermost state
- B) Same isotopes
- C) Same oxidation numbers

264. Which of these molecules is nonpolar? (283)

- A) HCN
- B) HCl
- C) H₂
- D) HBr
- AND NO

265. What is a tertiary carbon (1590)

- A) C bonded to three H atoms
- B) Third C of a carbonaceous chain
- C) C bonded to three other C atoms
- D) sp³ hybridized C
- E) sp² hybridized C

266. esters are formed by the reaction of a carboxylic acid and (1241)

- A) alcohol
- B) ethers
- C) hydrocarbons
- D) amines
- E) ketones

267. Indicate which of these is the name of Na_3PO_4 (638)

- A) sodium phosphite
- B) sodium triphosphite
- C) mixed oxide of sodium and sulfur
- D) sodium phosphate

268. Which of these has covalent bonds (570)

- A) CH_4
- B) BaO
- C) NaO

269. Which of these elements has the smallest atomic radius? (158)

- A) Ar
- B) Xe
- C) Fr
- D) Na
- E) Pb

270. What does the maximum energy point of the following graph indicate (reaction-energy coordinate graph)

(1334)

- A) transition state
- B) triple point
- C) critical point
- D) point of invariance
- E) melting points

271. Which aqueous solution among the following, with the same concentration, has the highest pH value: (1211)

- A) NaCl
- B) H_2CO_3
- C) NH_4Cl
- D) Na_2CO_3
- E) HCl

272. A resonance hybrid A, has two limit formulas A1 and A2. In what form can it be found in a composed? (374)

- A) Just like A
- B) Same as A1, A2 and A
- C) Same as A1 and A

273. Which of the following molecules is a peroxide (39)

- A) BaO₂
- B) H₂O
- C) CO₂
- D) Cl₂O₇

274. What does the calorimeter measure? (1246)

- A) Heat released during a reaction
- B) Absorbed heat
- C) Specific heat

275. One mole of lithium chloride is dissolved in 0.500 kg of water. What is the concentration of resulting solution? (351)

- A) 0.5 mol/l
- B) 0.5m
- C) 1 m
- D) 2M
- E) 1.5m

276. How many protons are contained in one mole of molecular hydrogen? (Na stands for $6.022 \cdot 10^{(23)}$) (421)

- A) 1 mole
- B) 2 moles
- C) $2/\text{Na}$
- D) $1 \cdot \text{Na}$
- E) $1 \cdot 6.022$

277. Which of these elements has the lowest first ionization energy (684)

- A) Fluorine
- B) Hydrogen
- C) Sodium
- D) Magnesium
- E) Oxygen

278. Ammonia is: (601)

- A) a strong foundation
- B) an acid of Arrhenius
- C) a Brønsted–Lowry basis
- D) a Brønsted–Lowry acid
- E) amphothera

279. Which of these statements is surely correct? (1546)

- A) They are all correct
- B) The equilibrium constant depends on the pressure
- C) The equilibrium constant depends on the concentrations
- D) The equilibrium constant depends on the volume
- E) The equilibrium constant depends on the temperature

280. A solution of a mixture has Van't Hoff coefficient equal to 3. Considering that is completely dissociated, what is the value of z ? (110)

- A) 3
- B) 2
- C) 4
- D) It depends on the concentration

281. Which of these is the best reducing agent (397)

- A) Br
- B) Cu^{2+}
- C) Pb(s)

282. What property is related to the secondary quantum number l ? (1360)

- A) Most of the energy of the orbital
- B) The orientation in space of the orbital
- C) The shape
- D) The spin
- E) The dimensions of the orbital

283. what happens when a solid melts (154)

- A) the Gibbs energy remains constant
- B) entropy increases
- C) the density decreases
- D) the temperature rises

284. Which statement is incorrect about alkaline earth metals (1140)

- A) They all react violently with water
- B) None of the above
- C) Their cations have the noble gas configuration
- D) At least one is radioactive
- E) They have two unpaired electrons

285. cyclohexane is the isomer of... (210)

- a) benzene
- B) hexane
- C) hexene

286. How many atoms are there in a central body cubic lattice? (1544)

- A) 1
- B) $1 + \frac{1}{8}$
- C) 2
- D) 4
- E) 9

287. The freezing point lowering of a certain aqueous solution is 1°C . What does it mean? (2097)

- A) At $+1^{\circ}\text{C}$ the first crystal of solution appears
- B) At -1°C the first crystal of solution appears
- C) At -1°C the solution is completely solid
- D) At $+1^{\circ}\text{C}$ the first crystal of pure ice appears
- E) At -1°C the first crystal of pure ice appears

288. Which compound is more acidic? (646)

- A) H_3PO_4
- B) H_2PO_4^-
- C) SH
- D) two other organic compounds

289. What is the geometry of the CO_2 molecule (243)

- A) Planar
- B) Trigonal
- C) Tetrahedral
- D) Linear
- E) Pyramidal

290. How many atoms does the face-centered cubic lattice contain? (1148)

- A) 1
- B) 2
- C) 6
- D) 5
- E) 4

291. Which of the following compounds is produced by the reaction between nitric acid and zinc hydroxide? (348)

- A) $\text{Zn}(\text{NO}_3)_2$
- B) ZnN_2
- C) Zn_2NO_3
- D) $\text{Zn}(\text{NO}_2)_2$
- E) Zn_2N

292. A definite amount of gas has (576)

- A) Defined mass
- B) Defined shape and volume
- C) Defined volume
- D) None of the others

293. In the case in which a reaction takes place at constant volume and constant T, it can be said that the heat released or absorbed and always (1563)

- A) Equal to the change in internal energy
- B) Equal to the enthalpy variation
- C) Dependent on system volume
- D) Independent of temperature
- E) Same as that released at constant pressure

294. CO_2 state diagram; the application asked for the state of aggregation under normal conditions (409)

- A) Solid
- B) Liquid
- C) Steam
- D) None of the above

295. For which of the following compounds can hydrogen bonding be predicted? (344)

- A) H₂S
- B) CO₂
- C) HF
- D) HCl
- E) HBR

296. What are the coefficients of the REAGENTS in the following reaction: Hydrogen gas + Nitrogen gas -> Ammonia (86)

- A) 3, 1
- B) 1, 3
- C) 2, 4
- D) 4, 2
- E) 1, 3, 4

297. A copper rod is immersed in a solution of zinc ions. Taking into account the potentials of standard reduction, it happens that: (83)

- A) Copper melts
- B) copper and zinc form an alloy of brass
- C) the zinc solidifies on the copper rod
- D) copper and zinc form a bronze alloy
- E) nothing happens

298. What cannot be an element with electron configuration 1 s² 2 s² 2 p⁶ (679)

- A) A pure transition metal
- B) No elements
- C) A pure noble gas
- D) A cation
- E) An anion

299. Which of the following elements has the highest first ionization energy (1741)

- A) F
- B) Ne
- C) St
- D) Cu
- E) Cs

300. Which compound does not follow Lewis's octet rule (I don't remember them all) (1691)

- A) PCl_5
- B) HCN
- C) NH_3

301. Which of the following compounds has a basic character? (1163)

- A) An amine
- B) An alcohol
- C) An alkane
- D) An alkene
- E) A ketone

302. Which of these is a tertiary amine? (1309)

- A) An amine functional group bonded to 6H
- B) An amine functional group bonded to 3H
- C) A functional group of amines bonded to 6C
- D) A functional group of amines bonded to 3C
- E) A functional group of amines with triple bond

303. What is the symbol of nitrogen? (213)

- A) A
- B) No
- C) O
- D) C
- E) Company

304. In the reaction : $\text{A} + 2\text{B} \rightarrow \text{C}$, 1 kg of A and 1 kg of B are reacted. What is the limiting reactant? [I don't remember all the possible answers] (1355)

- A) A
- B) B
- C) There is no limiting reactant
- D) It depends on the molar masses of A and B

305. Which of the following is Molecularly bonded? (1350)

- A) SiO_2
- B) Melted Iron
- C) HCl
- D) HF
- E) Frozen Water (Ice)

306. What represents the number of dissociated moles divided by the number of initial moles? (659)

- A) Degree of dissociation
- B) Dissociation constant
- C) Van 't Hoff coefficient

307. Ethylbenzene (1748)

- A) C₆H₁₀
- B) C₈H₁₀
- C) C₇H₁₂
- D) C₆H₁₂
- E) C₈H₁₂

308. Formula of Sodium Bicarbonate (Baking Sodium Acid) (1329)

- A) Na(CO)₃
- B) Na₂CO₃
- C) NaCO₂
- D) NaHCO₃
- E) NaH₂CO₃

309. The theory of ... states that acids are substances which transfer an H⁺ ion to another species

chemistry. (1357)

- A) Arrhenius
- B) Brownsted-Lowry
- C) Lewis

310. Complete the sentence: " Second _____ an acid is a substance which in aqueous solution increases the hydrogen ion concentration" (618)

- A) Bronsted Lowry
- B) Lewis
- C) Arrhenius
- D) Copenhagen
- E) Nernst

311. what is the atomic weight of boron-14 (isotope) (202)

- A) 10
- B) 14
- C) 10.81
- D) 5

312. During the discharge of the Daniel cell (637)

- A) The zinc electrode increases its mass
- B) The copper electrode increases its mass
- C) There is no effect
- D) The lead electrode increases its mass

313. Water temperature at its triple point (223)

- A) 0.01°C
- B) 0°C
- C) 101°C
- D) 100°C

314. Which of these properties is attributable to Benzene? (614)

- A) Its structural formula is C_6H_{12}
- B) Carbon is hybridized sp
- C) The bonds are saturated
- D) It is very responsive
- E) The benzene molecule is planar

315. Which of the following is a homogeneous mixture? (1785)

- A) I smoke
- B) Sand
- C) Metallic alloy
- D) Foam
- E) Fog

316. Daniel's stack is a stack (1133)

- A) Zinc - Carbon
- B) Zinc - Copper
- C) Alkaline

317. The electron configuration $[Kr] 4d^{10} 5s^0$ belongs to: (1194)

- A) It does not exist on the periodic table
- B) Silver
- C) Rhodium
- D) Palladium
- E) Cadmium

318. Which step makes it go from solid to gaseous without passing from the liquid phase? (1740)

- A) Perspiration
- B) It doesn't exist
- C) Sublimation
- D) Evaporation
- E) Merger

319. Balancing the following reaction: Barium nitrate + Calcium carbonate -> Calcium nitrate + Barium carbonate (703)

- A) 1 1 1 1
- B) 1 0.5 0.5 1
- C) 2 1 3 2
- D) 2 0.5 1 2

320. What is the formula of 2-propanol? (258)

- A) C₃H₈O
- B) C₃H₈
- C) C₃H₆
- D) C₃H₆O

321. What is the formula of cyclopentane? (683)

- A) C₅H₁₂
- B) C₆H₁₂O₆
- C) C₈H₁₅
- D) C₁₀H₅
- E) C₅H₁₀

322. What is the minimum amount of oxygen gas required to completely oxidize 46 g of sodium? (1646)

- A) 2 mol
- B) 23g
- C) 1 mol
- D) 32g
- E) 0.5 mol

323. What happens at the anode of $\text{Zn} + \text{HgO} \rightarrow \text{ZnO} + \text{Hg}$ (1345)

- A) $\text{Zn} + 2\text{e}^- \rightarrow \text{Zn}^{2+}$
- B) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
- C) $\text{HgO} + 2\text{e}^- \rightarrow \text{Hg}$
- D) $\text{Hg} + 2\text{e}^- \rightarrow \text{Hg}^{2+}$
- E) Nothing happens, the reaction does not produce current because it is not spontaneous

324. Which of the following molecules has cis and trans isomerism? (1205)

- A) 3-methyl-1-hexene
- B) 2-methyl-1-hexene
- C) 2-hexene
- D) 2-methyl-2-hexene
- E) 2-methyl-3-propyl-hexene

325. How many moles of magnesium oxide can be obtained with 1 mole of magnesium and 1 mole of oxygen (416)

- A) 1 mole
- B) 2 moles
- C) 1/2 mole
- D) no right answer
- E) 4 moles

326. Which of the following statements is correct (1746)

- A) There is only one type of intermolecular force
- B) There are multiple types of intermolecular forces
- C) Covalent intermolecular forces are the strongest

327. The cryoscopic constant... (1540)

- A) Together with the molality it influences the lowering of the freezing temperature
- B) Together with the molarity it influences the freezing point rise
- C) Together with the molality, it affects the freezing temperature rise
- D) Together with molarity it affects freezing point depression
- E) It is always in kJ

328. The sp^3 hybridization model can be used to represent the structure of : (1559)

- A) Acetylene C_2H_2 (flammable gas)
- B) ammonium cation (NH_4^+)
- C) C graphite (covalent solid)
- D) Benzene C_6H_6 (liquid)

329. Which of these is bromic acid? (385)

- A) HBR
- B) HBrO
- C) HBrO₂
- D) HBrO₃
- E) none of the above

330. Name of HClO₄ (251)

- To nobody
- B) Hydrochloric acid
- C) Hypochlorous acid
- D) Chlorous acid
- E) Chloric acid

331. Given Ag and Cu, which statement is true? (1744)

- A) Ag reduces faster than Cu
- B) Cu reduces faster than Ag
- C) Ag is the reducing agent
- D) Ag oxidizes faster than Cu
- E) None of these

332. Which of these has a carboxyl group (571)

- A) Alcohols
- B) Foreign
- C) Ethers

333. In an isothermal reaction, the product of volume and pressure (653)

- A) It was first discovered by Gay Lussac
- B) It is constant for ideal gases
- C) It is constant for any gas
- D) None of the above completely expresses the answer

334. What is C₆H₁₂? (562)

- A) cycloalkane
- B) alkane
- C) there is no unique answer
- D) alkene
- E) aromatic

- 335.** What is the relationship between the change in internal energy E and the enthalpy H for the reaction $A+B \rightarrow C$, carried out at a temperature of T Kelvin? (238)
- A) $E = -H$
 - B) $E = -H + RT$
 - C) $E = H + RT$
 - D) $E = H$
 - E) $E = H - RT$
- 336.** What is the change of state from solid to gas called? (177)
- A) Vaporization
 - B) Evaporation
 - C) Sublimation
 - D) It has no name
- 337.** Of which category of compounds is the $-OO-$ group typical? (270)
- A) Peroxides
 - B) Ethers
 - C) Foreign
- 338.** Calculate the Van't Hoff coefficient of magnesium acetate 2 M? (1245)
- A) I don't have enough data
 - B) 3
 - C) 4
 - D) 2
- 339.** In the nitric acid molecule, nitrogen has: (342)
- A) one simple and two double bonds.
 - B) the molecule is a resonance hybrid and the negative charge is distributed over three oxygen atoms.
 - C) two single bonds and one double bond.
 - D) three double bonds.
 - E) three simple bonds
- 340.** A gas is expanded (at constant T) from V_1 (with P_1) to twice the volume, what is the value of P ? (584)
- A) $P_2 = P_1$
 - B) $P_2 = 2 \cdot P_1$
 - C) $P_2 = P_1/2$
 - D) $P_2 = P_1/4$

341. What is the ratio between the dissociated moles and the total moles called? (676)

- A) degree of dissociation
- B) dissociation constant
- C) degree of solubility

342. Which of these elements has the greatest first ionization energy? (280)

- A) F
- B) Cn
- C) kr
- D) Approx
- E) Fe

343. Which contains ionic bonds? (1582)

- A) CaO
- B) CO₂
- C) H₂O
- D) Cl₂

344. What compound is obtained from the reaction of carbonic acid with magnesium hydroxide? (490)

- A) MnCO
- B) MnCO₃
- C) MgCO₃
- D) MgCO₂
- E) MgCO

345. the curve of the pressure vs volume graph for an isochoric transformation has a trend: (1358)

- A) Linear
- B) Quadratic
- C) Exponential
- D) Logarithmic
- E) Hyperbolic

346. Which of these liquid compounds can form a hydrogen bond? (399)

- A) HF
- B) H₂
- C) CH₄
- D) H₂S
- E) PO₃

347. What is the formula of 2-propanol (or propan-2-ol)? (1550)

- TO) **C3H7O**
B) **C3H8O2**
C) **C3H7O**
D) **C3H8O**
AND) **C3H8**

348. one mole of argon contains (211)

- A) 40 x Na
B) 40g
C) Avogadro's number of Ar atoms

349. What is the most stable allotrope of carbon at standard temperature and pressure? (1347)

- A) Graphite
B) Fullerene
C) Carbon nanotubes
D) Diamond
E) Amorphous carbon

350. Select the property which does not belong to ionic solids (1860)

- A) Fragility
B) High melting point
C) Good solid state electrical conductivity
D) Good electrical conductivity in the molten state
E) Good electrical conductivity in aqueous solutions

351. H₂S is a (401)

- A) Hydracid
B) Hydride
C) Salt

352. Mean molar mass of 10 vol% O₂ and 90 vol% H₂ (1538)

- A) 5 mol/L
B) 2.5 mol/L
C) 1 mol/L
D) 0 mol/L
E) 2 mol/L

353. select the exact quantity contained in 1 mole of CO (488)

- A) 32g of oxygen
- B) 1 mole of oxygen molecules
- C) Avogadro's number of colbate atoms
- D) 16g of oxygen
- E) Avogadro's number of oxygen molecules

354. (Image of a cubic unit cell with 8 particles at the ends) What kind of unit cell

And? (1672)

- A) Cubic with centered curve
- B) Face-centered cubic
- C) Simple cubic
- D) Hexagonal
- E) None of the above

355. What is the functional group of ketones? (149)

- A) -CH₃
- B) CH₄
- C) -CO
- D) -COOH

356. What was discovered first? (1765)

- A) electrons
- B) protons
- C) neutrons
- D) neutrinos
- E) nucleoli

357. Sodium perchlorate can be obtained by reacting (1206)

- A) NaOH + HClO₄
- B) NaOH + HClO₃
- C) NaOH + NaClO₂
- D) NaOCl + KCl
- E) H₂O₂ + HClO₄

358. For which of these values is a process NEVER spontaneous? (1310)

- A) $\Delta H > 0$ $\Delta S > 0$
- B) $\Delta H < 0$ $\Delta S > 0$
- C) $\Delta G < 0$
- D) $\Delta H > 0$ $\Delta S < 0$
- E) $\Delta H < 0$ $\Delta S < 0$

359. Which of these is a stronger acid? (619)

- A) HNO_3
- B) H_2SO_4
- C) H^+
- D) NaOH
- E) H_2SO_3

360. what is the van't hoff coefficient of sugar in aqueous solution (256)

- A) γ
- B) 0
- C) 1
- D) 2
- E) 4

361. What is the electronic configuration of Cobalt? (1153)

- A) $[\text{Ar}] 3d^{10} 4s^1$
- B) $[\text{Ne}] 3s^2 3p^4$
- C) $[\text{Ar}] 3d^7 4s^2$
- D) $[\text{Ar}] 4d^7 4s^2$
- E) $[\text{Kr}] 3d^7 4s^2$

362. Which of these is a transition metal (1530)

- A) Na
- B) Pb
- C) Mg
- D) Yes
- E) None of the above

363. The principle that explains the irreversibility of natural processes (376)

- A) Law 0 of thermodynamics
- B) First law of thermodynamics
- C) Second law of thermodynamics
- D) Third law of thermodynamics

364. Select the amount of lithium nitrate that contains exactly 42 g of nitrogen. (349)

- A) 2 formula units
- B) 1 amu
- C) 3 mol
- D) 70g
- E) a molecule

365. Given the reaction: $2A(\text{gas}) + B(\text{gas}) \rightleftharpoons 3C(\text{gas})$. At equilibrium, 3 moles of A, 2 moles of B develop one mole of C and the reaction pressure is 2 atm. Calculate K_p (645)

- A) 3/36
- B) 1/18
- C) 2/15
- D) There is not enough information to give the answer

366. How is the substance with the symbol CO indicated with the IUPAC nomenclature? (81)

- A) carbon monoxide
- B) carbon monoxide
- C) carbon monoxide
- D) carbon dioxide
- E) carbon dioxide

367. What affects the van't Hoff coefficient? (633)

- A) Solubility
- B) Dilution
- C) Solvent molar mass
- D) Solute molar mass
- E) Temperature

368. Which of these is an isomer of ethanol? (1401)

- A) hexane
- B) ethine
- C) none of the above
- D) hexanol

369. Benzene formula (55)

- A) C_6H_6
- B) $C_6H_{12}O_6$
- C) C_6H_{12}

370. What is the temperature of the triple point of CO_2 ? (1769)

- A) $-56.6^\circ C$
- B) $0.001^\circ C$
- C) -56.6 K
- D) $-0.001^\circ C$
- E) None of the above

371. $PCl_5 = PCl_3 + Cl_2$ is at equilibrium. What happens if the pressure decreases? (174)

- A) The equilibrium shifts to the right
- B) The balance shifts to the left
- C) The equilibrium constant increases
- D) The equilibrium constant decreases
- E) None of the above

372. A voltaic cell transforms (164)

- A) Chemical energy into electrical energy
- B) Electricity into thermal energy
- C) Thermal energy into electricity

373. Dissolve NH_3 in water. What is the ion dissociation constant of ammonia? (1796)

- A) $1.8 \cdot 10^5$
- B) $-1.8 \cdot 10^5$
- C) $1.8 \cdot 10^{-5}$
- D) $-1.8 \cdot 10^{-5}$
- E) None of the above

374. What is the osmotic pressure of a $NaCl$ solution of volume 2l, 1 M at a temperature T? (1400)

- A) $p = 2 \cdot 0.082 \cdot T$
- B) $p = 1 \cdot 0.082 \cdot T$
- C) $p = 1 \cdot 0.831 \cdot T$
- D) $p = 2 \cdot 0.831 \cdot T$

375. Which group does C_6H_{12} correspond to (593)

- A) Alkynes
- B) Alkenes
- C) Aromatic hydrocarbons
- D) Cycloalkanes
- E) Two of the answers are correct

376. What is the mole fraction of water in a solution with 1.2 moles of ethanol and 0.8 moles of water?

(93)

- A) 0.4
- B) 0.8
- C) 1

377. When ammonia is transformed into NH_4^+ it acts as: (1793)

- A base
- B) acid
- C) cation
- D) amphoteric
- E) anion

378. In which of these compounds does Cl have the oxidation number +3? (1157)

- A) Cl_2
- B) $NaClO_3$
- C) Cl_2O_5
- D) $NaClO$
- E) $NaClO_2$

379. The name of the following compound: HIO_4 (1361)

- A) Periodic acid
- B) Iodic acid
- C) Iodine acid
- D) Hypoiodous acid
- E) Ionic anhydride

380. A solution cannot be (1580)

- A) metastable
- B) saturated
- C) supersaturated
- D) solid
- E) polyphase

How many acid hydrogens are there in **CH₃CH₂COOH** **381.** (1528)

- A) 6
- B) 2
- C) 1
- D) 5

382. Which of these is not an energy? (424)

- A) Heat
- B) Pressure
- C) Electricity
- D) Light
- E) None of the above

383. Which of these is the commonly used symbol for entropy? (381)

- A) I think I was very lucky to have such a question
- B) St
- C) H
- D) (strange greek letter)
- E) G

384. What kind of hybridization is present in the central atom of C when it forms methane? (254)

- A) sp
- b) sp³
- C) sp²
- d) sp³d²
- E) None of these

385. Which has covalent bonds? (559)

- A) KBr
- B) CsF
- C) NaCl
- D) H₂O

386. In a non-deformable container there is a gas, what happens if I increase the temperature (155)

- A) the pressure remains constant
- B) the pressure-temperature ratio remains constant
- C) change the density
- D) none of the above is correct

387. Select Acid Salt (358)

- A) KH
- B) HCO_3^-
- C) H_2CO_3
- D) KHS

388. For $n = 6$, what is the maximum value that the magnetic quantum number can assume? (1846)

- A) +3
- B) +4
- C) +5
- D) +6

389. one mole of Li_2O contains (1149)

- A) 14 grams of lithium
- B) an Avogadro number of lithium molecules
- C) 7 grams of lithium
- D) an Avogadro number of oxygen molecules
- E) 14 grams of oxygen

390. What bond is made up of the gaseous nitrogen molecule? (1703)

- A) Single covalent bond
- B) Double covalent bond
- C) Triple covalent bond
- D) Ionic bond

391. What is the number of molecules in one mole of NaCl ? (1516)

- A) $1 \cdot \text{Na}$
- B) $2 \cdot \text{Na}$
- C) 58g
- D) 1
- E) 0

392. The partial P of a gas of a mixture depends on (133)

- A) T
- B) v
- C) Mole fraction

393. Which of these compounds contains ionic bonds? (347)

- A) CO
- B) F₂
- C) MgO
- D) HF
- E) H₂O

394. What substance is more miscible in water ?? (649)

- A) ethanol
- B) propanol
- C) cyclopentane
- D) cyclohexane
- E) ethine

395. With what instrument is the chaos of a system measured? (1403)

- A) Libra
- B) Thermometer
- C) Barometer

396. The appropriate adjective for a reaction with positive enthalpy is: (45)

- A) Isochora
- B) Spontaneous
- C) Exothermic
- D) Endothermic

397. given the standard potential of Rubidium $E^\circ = -3V$ and of Beryllium $E^\circ = -2V$, calculate the standard potential of reaction (1239)

- A) -1
- B) 1
- C) -8
- Q) -4
- E) 4

398. Hess' law (406)

- A) He says that entropy is the maximum disorder
- B) He says that entropy is a state function
- C) None
- D) It allows to calculate the values of the electric potentials

399. According to the Lewis structure, how many valence electrons does nitrogen have in its ground state? (1216)

- A) 4
- B) 6
- C) 5
- D) 10

400. A solution is acidic if (644)

- A) $[\text{H}_3\text{O}^+] > 10^{-7}$
- B) $[\text{OH}^-] < 10^{-7}$
- C) $\text{pH} > \text{pOH}$
- D) $K_W < 10^{-7}$
- E) $[\text{OH}^-] > 10^{-7}$

401. Which of these is a molecule (574)

- A) Ice
- B) SiO_2
- C) Na
- D) NaCl
- E) Molten iron

402. Indicate, from left to right, the oxidation number assumed by sulfur in the following compounds:

Strontium sulphate, (another compound that I can't remember in which the oxidation number of sulfur is 4), SO_2 , SO_3 (103)

- A) 6,4,4,6
- B) 4,4,4,6
- C) 4,4,6,6
- D) 4,6,6,4
- E) 6,4,4,4

403. What is the amount of atomic sulfur required to form 64 g of SO₂ (1736)

- A) 1 mole
- B) 64 amu
- C) 64g
- D) 0.5 moles
- E) 48g

404. Which group does RCOOR' belong to (402)

- A) Foreign
- B) Peroxides
- C) Ethers

405. Which of the following choices has the greatest bond strength in descending order? (159)

- A) HI-HBr - HCl-HF
- B) HF- HCl- HBr - HI
- C) HCl-HF-HI-HBr
- D) HI-HF - HCl-HBr

406. Which of these factors does not affect bond strength: (639)

- A) The Electronegativity Difference
- B) The binding order
- C) The atomic radius
- D) All of these factors affect bond strength
- E) Bond length

407. Which compound has the formula C_nH_{2n+2} (29)

- A) Alkane
- B) Alkene
- C) Alkyne
- D) Benzene

408. Which of these has a link OO (1534)

- A) Alkynes
- B) Peroxides
- C) Carbonic acids

409. Select the electron configuration of the sulfide ion. (346)

- A) [Ne] 2s² 2p⁶
- B) 1s² 2s² 2p⁶ 3s² 3p⁶
- C) [Ne] 3s² 3p⁵
- D) 1s² 2s² 2p⁶ 3s⁰ 3p⁴
- E) 1s² 2s² 2p⁶ 3p⁴

410. What is the IUPAC name of acetylene (1719)

- A) Propane
- B) Ethene
- C) Propene
- D) Etino

411. the curve of the pressure vs volume graph for an isothermal transformation has a trend: (234)

- A) quadratic
- B) linear
- C) exponential
- D) logarithmic
- E) hyperbolic

412. Why does hydrogen have a standard reduction potential equal to 0? (690)

- A) It has the same tendency to oxidize and shrink
- B) is the first element of the table
- C) was chosen as the standard potential
- D) is the strongest oxidant

413. What does one mole of Argon correspond to? (415)

- A) 40kg
- B) $40 \times 6.022 \times 10^{23}$ g
- C) an avogadro number of particles
- D) 6.022×10^{23} g

414. in a lead battery discharge reaction which element is oxidized? (reaction was given)

(441)

- A) PbO₂
- B) Pb
- C) none

415. Which of these is a homogeneous mixture (316)

- A) fog
- b) smoke
- C) sand
- D) foam
- E) bronze

416. Which of the following statements refers to an endothermic reaction that takes place at temperature e constant pressure? (1518)

- A) The products have an enthalpy higher than that of the reactants and a positive ΔH
- B) The products have an enthalpy higher than that of the reactants and a negative ΔH
- C) The products have an entropy similar to that of the reactants
- D) The products have a lower enthalpy than that of the reactants and a higher ΔH
- E) The products have an enthalpy lower than that of the reactants and a lower ΔH

417. In a reaction we have $\Delta H = 2000 \text{ kJ}$ and $\Delta S = 20 \text{ kJ/K}$. How must the temperature be for the Is the reaction spontaneous? (642)

- A) $T > 100 \text{ K}$
- B) $T < 100 \text{ K}$
- C) There are not enough elements to answer
- D) For any temperature

418. How many electrons can occupy the same orbital at the same time? (14)

- A) 2
- B) 1
- C) 10
- D) 8
- E) 6

419. Hydrogen sulphide (1763)

- A) H_2S
- B) HS
- C) HSO_3
- D) HSO_4

420. Ethylene is a... (1858)

- A) alkene
- B) carboxylic acid
- C) alcohol
- D) alkane
- E) alkyne

421. Which of these is not a resonance hybrid (435)

- A) HNO_3
- B) SO_3
- C) NO_3^-
- D) CH_3COOH

422. Balance the following reaction: $\text{S}(2) + \text{molecular oxygen} \rightarrow \text{SO}(3)$ (648)

- A) 2:1
- B) 1:2
- C) 1:1
- D) 3:1
- E) 1:3

423. Which compound reacts with HCl and leads to the formation of chlorobutane? (245)

- A) Butanone
- B) Butanal
- C) Butane
- D) Butene
- E) Butanol

424. how many isomers of dibromobenzene? (1155)

- A) 2
- B) 3
- C) 4
- D) 6
- E) 0

425. The Van't Hoff coefficient depends on : (581)

- A) by the temperature
- B) by the solubility
- C) from the moles of solute
- D) from the moles of solvent

426. The solidification of water is a process: (99)

- A) exothermic
- B) Isentropic
- C) Isoenthalpic
- D) Favored at low pressures
- E) None of the above

427. Which of these is a nonmetal (semimetal)? (1778)

- A) Hydrogen
- B) Samarium
- C) Technetium
- D) Indium
- E) Antimony

428. Equal volumes of different gases, under the same conditions of pressure and temperature, contain the same number of molecules. What gas law is it? (94)

- A) Avogadro
- B) Lavoiser
- C) Gay Lussac
- Q) Boyle

429. Which of these families of compounds has the formula: $C_nH_{(2n+2)}$ (640)

- A) Alkynes
- B) Aldehydes
- C) Alkenes
- D) Alkanes
- E) Carboxylic acids

430. How many covalent bonds can Xe make by respecting the octet rule? (586)

- A) 2
- B) 1
- C) 3
- D) 4

431. At 1000K what is the relationship between K_c and K_p , if $\Delta n = 1$? (430)

- A) $K_p = 82 K_c$
- B) $K_p = K_c \cdot 82^2$
- C) $K_p = K_c/82$
- D) $K_p = K_c$

432. What is the formula of bromic acid (1754)

- A) HBrO
- B) HBrO_3
- C) HBrO_4
- D) HBr_2O
- E) H_2BrO_2

433. Which of these is an amine? (1859)

- A) CH_3CONH_2
- B) $\text{C}_6\text{H}_5\text{NH}_2$
- C) $\text{C}_6\text{H}_5\text{NO}$
- D) CCl_4

434. Which of these compounds contains an ionic bond (222)

- A) BaCl_2
- B) H_2
- C) CO_2
- D) O_2

435. Which of these principles explains irreversibility [...] ? (1589)

- A) First principle
- B) Second principle
- C) Third principle
- D) Lavoisier's principle

436. Which of these compounds is a peroxide? (361)

- A) H_2O
- B) Na_2O
- C) Cl_2O_7
- D) CO_2
- E) BaO_2

437. In a cell a copper plate is immersed in a solution of zinc ions. What happen? (1142)

- A) Copper is covered with a layer of zinc
- B) The reaction does not take place
- C) Copper dissolves
- D) Copper and zinc react to form a bronze alloy
- E) Copper and zinc react to form a silver alloy

438. The sublimation of an ice cube is a process: (1783)

- A) Exothermic
- B) Endothermic
- C) Spontaneous
- D) None are correct

439. Fe₂O₃ nomenclature (1551)

- A) Ferric oxide
- B) Ferrous oxide
- C) Iron (I) oxide
- D) Iron(II) oxide

440. Which of the following ions is isoelectronic with Argon? (1200)

- A) F
- B) P³⁺
- C) Rb⁺
- D) S²⁻
- E) Al³⁺

441. how many molecules are contained in 14 moles of H₂S (493)

- A) 14*6.022
- B) Avogadro's number
- C) 14
- D) 14 divided by Avogadro's number
- E) 14 * avogadro's number

442. Which of these statements is NOT valid for ethylene? (437)

- A) polymers can be obtained from it
- B) is a saturated hydrocarbon
- C) from addition reactions

443. What is the letter commonly used to indicate entropy (122)

- A) H
- B) St

THERE IS

444. -COOH What compounds are characterized by this group? (184)

- A) Carboxylic acids
- B) Alkenes
- C) Aldehydes

445. At the same pressure and temperature, one mole of He gas occupies a volume equal to that occupied by : (350)

- A) 16 g of oxygen
- B) two moles of a monatomic gas
- C) half a mole of a diatomic gas
- D) 14 g of nitrogen
- E) 28 g of nitrogen

446. Consider evaporating water. Which of these statements is definitely correct? (239)

- A) The temperature is 100°C
- B) All other statements are wrong
- C) Entropy increases
- D) The vapor pressure of water varies

447. The correct formula of the van't Hoff coefficient is: (1801)

- A) $i = 1 - z(a+1)$
- B) $i = 1 + z(a-1)$
- C) $i = 1 + a(z-1)$
- D) $i = 1 - a(z-1)$
- E) None of the above

448. What is the structural formula of fluorine gas? (7)

- A) FF
- B) F=F
- C) F $\ddot{y}$ F
- D) F
- E) FFF

449. What is the correct cell notation of Daniell's pile? (677)

- A) $\text{Zn}^{2+} \mid \text{Zn} \parallel \text{Cu} \mid \text{Cu}^{2+}$
- B) $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$
- C) $\text{Cu} \mid \text{Cu}^{2+} \parallel \text{Zn}^{2+} \mid \text{zn}$
- D) $\text{Cu} \mid \text{Cu}^{2+} \parallel \text{Zn} \mid \text{Zn}^{2+}$
- E) $\text{Cu} \mid \text{Cu}^{2+} \parallel \text{Zn}^{2+} \mid \text{zn}$

450. What is the formula of sodium monoacid phosphate? (652)

- A) Na_2HPO_4
- B) Na_2HPO_3
- C) NaH_2PO_4
- D) NaH_2PO_3

451. Calculate how many molecules are present in 40 g of helium? (1669)

- A) 40 Na
- B) 10 Na
- C) 10/Na
- D) 5 Na

452. What are the instantaneous dipole-induced dipole intermolecular forces called (411)

- A) London forces
- B) Forces of Paris
- C) Copenhagen Forces
- D) Tokyo Forces

453. What is the formula of potassium permanganate (1333)

- A) K_2MnO_4
- B) KMnO_4
- C) K_2MnO_4
- D) K_2MnO_3
- E) KMnO_3

454. Oxidation number of sulfur in silver sulfide (605)

- A) 2
- B) 4
- C) -2
- D) 6
- E) 5

455. What does the 12 in carbon-12 represent? (1777)

- A) Atomic number
- B) Avogadro's number
- C) Mass number
- D) Atomic weight
- E) Number of electrons

456. Represents Daniell pile pattern (1151)

- A) $\text{Zn}|\text{Zn}^{2+}||\text{Cu}^{2+}|\text{Cu}$
- B) $\text{Cu}^{2+}|\text{Cu}||\text{Zn}|\text{Zn}^{2+}$
- C) $\text{Cu}^{2+}|\text{Zn}||\text{Cu}|\text{Zn}^{2+}$
- Q) $\text{Zn}|\text{Cu}||\text{Zn}^{2+}||\text{Cu}^{2+}$

457. What is the maximum value of the quantum number n ? (478)

- A) 6
- B) 7
- C) 5
- D) Infinity
- E) 1

458. which of these is not a resonance hybrid (188)

- A) SO_2
- B) CCl_4
- C) C_6H_6
- D) they are all resonance hybrids
- E) O_3

459. Which compound is more reactive? (667)

- A) Acetylene
- B) Methane
- C) Benzene
- D) Cyclohexane
- E) Propane

460. In an insulated container there are two identical glasses. One contains 100ml of pure water and the other contains 100ml of sugar water. At equilibrium which glass will have the most liquid (575)

- A) The one with pure water
- B) The one with water and sugar
- C) They are the same

461. Which pair of substances is not miscible? (1204)

- A) $\text{CH}_3\text{-OH}$ - H_2O
- B) H_2SO_4 - H_2O
- C) C_6H_{12} - H_2O
- D) NaCl - H_2O
- E) NH_3 - H_2O

- 462.** For a certain reaction $\Delta H = 2000 \text{ kJ}$ $\Delta S = 20 \text{ kJ/K}$. Select the appropriate temperature range in which this reaction occurs spontaneously if the pressure does not change. (1648)
- A) The reaction is never spontaneous
 - B) The reaction is spontaneous at any temperature
 - C) $T < 100\text{K}$
 - D) $T > 100\text{K}$
 - E) lack of data to be able to answer

- 463.** Pressure and temperature under standard conditions (699)
- A) 1 bar and 25°C
 - B) 1 atm and 0°C
 - C) 1 atm and 25°C

Balance the reaction $\text{SO}_2 + \text{O}_3 \rightarrow \text{SO}_3$ **464.** (617)

- A) 3 1 3
 - B) 2 1 2
 - C) 1 1 1
 - D) 1 0.5 0.5
- 465.** If I increase the pressure I go from a liquid state to a solid state, is this statement true? (1402)
- A) Under no circumstances
 - B) always true
 - C) true for most substances
- 466.** Which of these compounds can form a hydrogen bond in the liquid state? (613)
- A) CH_3OH
 - B) NH_3
 - C) CH_4
 - D) H_2S
 - E) none of the above
- 467.** if I have a non-deformable container with ideal gas inside at a pressure of 1 atm at 1°C . If the pressure is raised to 2 atm what is the temperature? (433)
- A) 275°C
 - B) 275K
 - C) 2°C
 - D) None of the others

468. Which of the following molecules is apolar? (244)

- A) HI
- B) Br₂
- C) HCl
- D) HCN

469. If, under the same conditions, fluid A has a higher vapor pressure than fluid B then: (1404)

- A) B is more viscous than A
- B) A is denser than B
- C) B is denser than A
- D) A is more volatile than B
- E) A is more viscous than B

470. If N_A = Avogadro's number, 0.4 moles of hydrogen sulphide correspond to (1529)

- A) $(0.4 \times 2)/N_A$ hydrogen atoms and $0.4/N_A$ sulfur atoms
- B) 0.4 liters of hydrogen sulfide under normal conditions
- C) $(0.4 \times 2 \times N_A)$ hydrogen atoms and $(0.4 \times N_A)$ sulfur atoms
- D) $(0.4 \times N_A)$ hydrogen atoms and $(0.4 \times N_A)$ sulfur atoms
- E) 0.4 liters of hydrogen sulphide at 25°C and 1 atm

471. Which of these factors does not affect the rate of a reaction? (27)

- A) Temperature
- B) Catalyst
- C) Activation energy
- D) Number of collisions
- E) None of the above

472. How are hydroxides formed? (643)

- A) Water + non-metal
- B) Metal + oxygen
- C) Metal + water
- D) Basic oxide + water
- E) Acid oxide + water

473. From which of the following mixtures does hydrogen evolve? (1207)

- A) $\text{Cu} + \text{H}_2\text{SO}_4$
- B) $\text{Ag} + \text{HCl}$
- C) $\text{Cl} + \text{H}_2\text{SO}_4$
- D) $\text{Pt} + \text{HCl}$
- E) $\text{Zn} + \text{HCl}$

474. Where does reduction take place in a galvanic cell? (682)

- A) anode
- B) cathode
- C) nowhere
- D) in the salt bridge
- E) It's everywhere

475. Unit of measurement of entropy? (1182)

- A) $1/\text{J}$
- B) J
- C) $\text{J} \cdot \text{K}$
- D) J/K
- E) It has no units of measure

476. What is a hydride? (678)

- A) Any hydracid
- B) Any compound with H
- C) Substance where H has oxidation number -1
- D) An acid

477. How is sulfuric acid formed by reacting one of these components with water? (1742)

- A) SO_2
- B) SO_3
- C) SO_4^{2-}
- D) SO
- E) None of these

478. What is the formula of benzene? (1720)

- A) C_6H_{12}
- B) C_8H_8
- C) C_6H_8
- D) C_6H_6
- E) C_6H_4

479. When bonding occurs, the process may be defined (1747)

- A) Always spontaneous
- B) Never spontaneous
- C) None of these

480. How many moles of products does the combustion of 2 moles of methane produce? (585)

- A) 4
- B) 6
- C) 8
- D) 10

481. Water begins to boil when... (432)

- A) the vapor pressure exceeds the external pressure
- B) the vapor pressure equals the external pressure
- C) always at 100 degrees without exception
- D) when some particles begin to evaporate

482. In the electrolysis of an aqueous solution of KCl, at the cathode we obtain: (1208)

- A) Molecular hydrogen
- B) Gaseous molecular oxygen
- C) Metallic potassium
- D) Atomic oxygen

483. What are actinides or actinides? (I don't remember the 5th answer) (1721)

- A) Elements of the 9th period
- B) elements of the 7th period that fill the 7f
- C) elements of the 7th period that fill the 6f
- D) elements of the 7th period that fill the 5f

484. There is a theory that an acid is a species which: (1543)

- A) Accept an electron
- B) Donate an electron
- C) Accept a proton
- D) Donate a proton
- E) Reacts with water

485. select the correct statement regarding methanol (480)

- A) contains a sp hybridized carbon
- B) its molecular formula is CH₂O
- C) its molecular mass is 32
- D) contains a carboxyl group
- E) is an aromatic compound

486. One mole of nitrogen gas and one mole of oxygen gas have the same temperature.

mean speeds of their molecules it can be said that: (1522)

- A) It is greater for nitrogen
- B) They are exactly the same
- C) It is greater for oxygen
- D) They are proportional to the square root of the molecular weights
- E) None of the other answers

487. What is nitrate? (1343)

- A) NO₃⁻
- B) NO₄⁻
- C) NH₄
- D) NH₃
- E) An atom

488. The formula C_nH_{2n}, which class of compounds does it represent? (1784)

- A) Alkynes
- B) Alkenes
- C) Alkanes
- D) Aromatic hydrocarbons
- E) Cycloalkynes

489. What is the transition from solid state to vapor state called? (611)

- A) sublimation
- B) evaporation
- C) fusion
- D) vaporization
- D) condensation

490. What is this organic molecule called? Image of C₂H₄ (double bond between the two Cs) (47)

- A) Ethylene
- B) Ethane
- C) Acetylene
- D) Diet

491. Balance the following reaction and write the stoichiometric coefficients of the reactants: Fe(OH)₂ + K₂CrO₄ + H₂O --> Fe(OH)₃ + KCr(OH)₄ + KOH (1199)

- A) 3 1 4
- B) 3 3 4
- C) 2 1 4
- D) 3 2 4
- E) 1 1 4

492. What does the lowering of steam pressure depend on? (1344)

- A) molality
- B) molarity
- C) temperature
- D) volume
- E) mole fraction

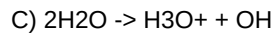
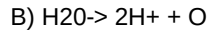
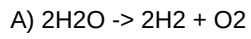
493. How many liters do 16 g of oxygen occupy at 0 °C and 1 atm (187)

- A) 22.4
- B) 11.2
- C) 36
- D) 24.2

494. How many covalent bonds does Xe form to complete the octet? (1733)

- A) 1
- B) 2
- C) 3
- D) 4

495. Electrolysis reaction of water. (688)



496. van't hoff coefficient (324)

A) $i = 1 - \alpha(z+1)$

B) $i = 1 - \alpha(z-1)$

C) $i = 1 + \alpha(z-1)$

D) $i = 1 + \alpha(z+1)$

Solutions:

Verified quizzes are underlined

<u>1 A</u>	<u>2c</u>	<u>3C</u>	<u>4 E</u>	<u>5c</u>	<u>6B</u>	<u>7 A</u>	<u>8 c</u>	<u>9 E</u>	<u>10 D</u>
<u>11 to 12 to 13 C</u>	_____	<u>14b</u>	<u>15D 16A 17A 18A 19A 20B</u>	_____	_____	_____	_____	_____	_____
<u>21b</u>	<u>22A 23A 24A 25C</u>	_____	_____	<u>26A 27D 28B</u>	_____	<u>29b</u>	<u>30A</u>	_____	_____
<u>31b</u>	<u>32D 33D 34B</u>	_____	<u>35B</u>	<u>36 C</u>	<u>37 C</u>	<u>38B</u>	<u>39 to 40 d</u>	_____	_____
<u>41 C</u>	<u>42D 43C</u>	_____	<u>44D 45A 46A 47D 48B</u>	_____	_____	_____	<u>49A 50A</u>	_____	_____
<u>51B</u>	<u>52D 53B</u>	_____	<u>54A 55B</u>	_____	<u>56A 57A 58A 59D 60A</u>	_____	_____	_____	_____
<u>61 C</u>	<u>62b</u>	<u>63 E</u>	<u>64B</u>	<u>65D 66A 67A 68D 69A 70C</u>	_____	_____	_____	_____	_____
<u>71A 72B</u>	_____	<u>73D 74C</u>	_____	<u>75D 76A 77E</u>	_____	<u>78D 79C</u>	_____	<u>80 D</u>	_____
<u>81 C</u>	<u>82A 83A 84B</u>	_____	<u>85B</u>	<u>86 E</u>	<u>87B 88A 89A 90E</u>	_____	_____	_____	_____
<u>91 c</u>	<u>92c</u>	<u>93B</u>	<u>94B</u>	<u>95B</u>	<u>96 D 97 C</u>	_____	<u>98D 99D 100C</u>	_____	_____
<u>101A 102E</u>	_____	<u>103C 104B 105D 106C 107C 108A 109A 110A</u>	_____	_____	_____	_____	_____	_____	_____
<u>111E 112B 113A 114A 115A 116B 117C 118D 119A 120D</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>121C 122D 123B 124B 125E 126D 127E</u>	_____	_____	_____	_____	_____	<u>128C 129E 130B</u>	_____	_____	_____
<u>131A 132C 133C 134C 135C 136A 137B 138A 139A 140D</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>141C 142C 143A 144C 145D 146B 147B 148A 149C 150C</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>151A 152C 153A 154A 155C 156C 157B 158B 159A 160D</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>161A 162D 163A 164E 165B 166A 167A 168B 169A 170A</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>171B 172A 173C 174A 175C 176A 177A 178A 179D 180A</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>181A 182A 183D 184C 185A 186D 187B 188A 189C 190B</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>191A 192B 193A 194C 195B 196A 197A 198B 199D 200A</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>201A 202C 203A 204A 205D 206D 207E</u>	_____	_____	_____	_____	_____	<u>208C 209E 210C</u>	_____	_____	_____
<u>211C 212E</u>	_____	<u>213D 214D 215B 216C 217D 218A 219B 220E</u>	_____	_____	_____	_____	_____	_____	_____
<u>221A 222C 223A 224D 225B 226C 227A 228A 229C 230C</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____
<u>231B 232C 233E 234A 235B 236C 237C 238A 239B 240A</u>	_____	_____	_____	_____	_____	_____	_____	_____	_____

241A 242C 243B 244D 245B 246C 247A 248D 249D 250A _ _ _ _ _
251 C 252 E _ _ _ 253E 254B 255B 256A 257C 258B 259B 260C _ _ _ _ _
261C 262D 263A 264C 265C 266A 267D 268A 269A 270A _ _ _ _ _
271D 272A 273A 274C 275D 276B 277C 278C 279E 280A _ _ _ _ _
281C 282C 283B 284A 285C 286C 287E _ _ _ _ _ 288A 289D 290E _ _ _
291A 292A 293A 294C 295C 296A 297E _ _ _ _ _ 298A 299B 300A _ _ _
301A 302D 303B 304D 305E 306A 307B 308D 309B 310C _ _ _ _ _
311B 312B 313A 314E 315C 316B 317D 318C 319A 320A _ _ _ _ _
321 and 322 e _ _ _ 323B 324C 325A 326B 327A 328B 329D 330A _ _ _ _ _
331A 332B 333B 334C 335C 336C 337A 338B 339C 340C _ _ _ _ _
341A 342A 343A 344C 345A 346A 347D 348C 349A 350C _ _ _ _ _
351A 352A 353D 354C 355C 356B 357A 358D 359B 360C _ _ _ _ _
361 C 362 E _ _ _ 363C 364C 365B 366B 367D 368C 369A 370A _ _ _ _ _
371A 372A 373C 374A 375E 376A 377B 378E 379A 380E _ _ _ _ _
381C 382B 383B 384B 385D 386B 387D 388C 389A 390C _ _ _ _ _
391E 392C 393C 394A 395B 396D 397B 398C 399C 400A _ _ _ _ _
401A 402A 403A 404A 405B 406D 407A 408B 409B 410D _ _ _ _ _
411E 412C 413C 414B 415E 416A 417A 418A 419A 420A _ _ _ _ _
421 D 422 E _ _ _ 423E 424B 425C 426A 427E _ _ _ _ _ 428A 429D 430A _ _ _
431A 432B 433A 434A 435B 436E 437B 438B 439A 440D _ _ _ _ _
441 E 442 B 443 B 444 A 445 E 446 C 447 C 448 A 449 B 450 A _ _ _ _ _
451B 452A 453B 454C 455C 456A 457B 458B 459A 460B _ _ _ _ _
461C 462D 463A 464A 465C 466B 467A 468B 469D 470C _ _ _ _ _
471 E 472 D 473 E 474 B 475 D 476 C 477 B 478 D 479 C 480 B _ _ _ _ _
481B 482A 483D 484D 485C 486A 487A 488B 489A 490A _ _ _ _ _
491A 492E _ _ _ 493B 494A 495A 496C _ _ _ _ _